Foundations of Risk Management 目录

Quantitative Analysis 目录

Financial Markets and Products 目录

Valuation and Risk Models 目录

1. Case studies that feature financial engineering by way of complex derivatives include Bankers Trust, the Orange County case, and Sachsen Landesbank. In regard to these financial engineering cases, each of the following statements is true EXCEPT which is false?
通过复杂衍生品进行金融工程的案例研究包括 Bankers Trust、橙县案例以及萨克森州银行（Sachsen Landesbank）。关于这些金融工程案例，下列各项陈述均为真，除了哪一项为假?

A. Bankers Trust (BT) proposed an overly complex swap to their clients (P&G and Gibson Greetings) but the swaps experienced colossal losses; the clients sued BT, who never recovered from the ensuing reputational damage
Bankers Trust（BT）向其客户（宝洁 P&G 和 Gibson Greetings）提出了过于复杂的互换方案，但这些互换产生了巨额亏损；客户起诉了 BT，而 BT 也从未从随后造成的声誉损害中恢复

B. Orange County's treasurer (Robert Citron) borrowed through the repo market to purchase inverse floating-rate notes – positions that Citron later said he did not understand – but the combination of excessive leverage and embedded interest-rate risk generated losses that ultimately forced Orange County to file for bankruptcy
橙县财政官（Robert Citron）通过回购市场（repo market）借款，用于购买反向浮动利率票据——Citron 事后表示自己并不了解这些头寸——但过度杠杆与内嵌利率风险的组合最终导致巨额亏损，迫使橙县不得不申请破产

C. Sachsen Landesbank set up off-balance-sheet vehicles (guaranteed by Sachsen) that held highly-rated U.S. mortgage-backed securities; the operation was highly profitable, but the 2007-08 subprime crisis wiped out Sachsen's capital
萨克森州银行（Sachsen Landesbank）设立了由萨克森州银行提供担保的表外工具，这些工具持有高评级的美国抵押贷款支持证券；该业务在一段时间内利润颇丰，但 2007–08 年的次贷危机却抹去了萨克森州银行的全部资本

D. If the purpose of the position is designated as hedging (rather than speculation) and if the hedge consists only of some combination(s) of forwards, swaps and/or options – which are the primary building blocks – then the firm can avoid problems suffered by the financial engineering case studies because the firm avoids undue sophistication
如果某一头寸的目的被指定为套期保值（而非投机），且该套保仅由远期、互换和/或期权——这些主要构件——的某种组合构成，那么公司就可以避免这些金融工程案例中遭遇的问题，因为公司避免了过度复杂化

答案:D
解析:
D 选项错误。

- A hedge does not avoid the assumption of embedded risks that may not be understood; the hedge/speculation intention is important but does not completely warranty the risks of the position. More importantly, complex positions are combinations of the primary building blocks such that complex hedge positions do not avoid running significant

考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]
科目: Valuation and Risk Models

2. A very small sample of 10 public companies has one target variable (i.e., Dividend) and four features (i.e., Earnings, LargeCap, Retail, and Tech). Both Dividend and Earnings are binary variables where respectively, one (1) indicates the company recently paid a dividend, and its earnings dropped in the previous year. The data frame is displayed below.

Dividend	Earnings	LargeCap	Retail (%)	Tech
1	0	0	80	1
1	0	1	40	1
1	0	1	20	0
1	0	1	60	0
0	1	0	30	1
0	1	0	40	0
0	1	1	40	0
1	1	1	30	0
1	1	1	20	0
1	1	1	20	1

Based on Dividend as the output (aka, target) variable, the base Gini coefficient is 0.420. That's because $1-[(7/10)^2+(3/10)^2]$. What is the information gain if we split (as the root node's feature) the Earnings feature?

一组仅包含 10 家上市公司的极小样本具有一个目标变量（即 Dividend）和四个特征（即 Earnings、LargeCap、Retail 和 Tech）。Dividend 和 Earnings 均为二元变量，其中取值为 1 表示公司最近支付了股利，且其上一年的盈利出现了下滑。数据框如下所示。

Dividend	Earnings	LargeCap	Retail (%)	Tech
1	0	0	80	1
1	0	1	40	1
1	0	1	20	0
1	0	1	60	0
0	1	0	30	1
0	1	0	40	0
0	1	1	40	0
1	1	1	30	0
1	1	1	20	0
1	1	1	20	1

以 Dividend 作为输出（即目标）变量时，其基础 Gini 系数为 0.420。这是因为 $1 - [(7/10)^2 + (3/10)^2]$。如果我们以 Earnings 作为根节点的特征进行划分，其信息增益是多少?

A. Zero

B. 0.120

C. 0.348

D. 0.875

答案:B

解析:

B 选项正确。

- We need the weighted Gini coefficient for (splitting on) the Earnings feature. The four Earnings=0 all pay dividends: this branch is pure with a Gini of zero! The Gini of the six Earnings = 1 branch, where 3 pay dividends and 3 do not, is given by $1–(3/6)^2–(3/6)^2 = 0.50$. The weighted Gini is, therefore, 0*4/10 + 0.50*6/10 = 0.30. The information gain is 0.120 because splitting on the Earnings feature reduces from the base Gini of 0.420 to 0.300.

考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]

科目: Financial Markets and Products

3. About swaps, each of the following is true EXCEPT which is false?

关于互换（swaps），下列各项陈述均为真，除了哪一项为假?

A. At any given time during the life of a bilateral OTC swap, at least one of the counterparties incurs both market and credit risk
在一笔双边场外（OTC）互换的存续期内的任意时点，至少有一方交易对手承担市场风险和信用风险

B. A difference between the typical interest rate swap and currency swap is whether the principal (aka, notional principal) is exchanged
典型利率互换与货币互换之间的一个区别在于是否交换本金（即名义本金）

C. Both overnight indexed swaps (OIS) and volatility swaps need to determine a realized (aka, actual) financial variable at the end of a period or tenor
隔夜指数互换（OIS）和波动率互换都需要在某一期间或期限结束时确定一个已实现（即实际的）金融变量

D. A commercial bank funded by short-term floating-rate deposits that extends longterm fixed-rate loans can hedge market risk by entering a swap where it is the floating-rate payer and fixed-rate receiver
一家由短期浮动利率存款融资、却发放长期固定利率贷款的商业银行，可以通过进入一笔自身支付浮动利率、收取固定利率的互换来对冲市场风险

答案:D
解析:
D 选项正确。

- This is the opposite of a hedge! The bank hedges with a swap where it is the floating-rate receiver and fixed-rate payer In regard to (A), (B), and (C), each is TRUE

考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]
科目: Financial Markets and Products

4. A large international bank uses EWMA for their risk system, with a decay factor of 0.92. On day T the forecast volatility for two variables, X&Y, are identical at 1.25 and the current correlation forecast is 15.0. If on day T the realized returns for X & Y are 0.50 and -0.50 respectively, which of the following is closest to the new correlation?

A. 12.5%

B. 13.4%

C. 15.0%

D. 16.2%

答案:B
解析:
A 选项错误。

- 12.5% is the answer if we use yesterday's volatilities in the correlation equation

B 选项错误。

- Recall the EWMA equations for variance and covariance:

$$\sigma_{T+1}^2 = \lambda\sigma_T^2 + (1-\lambda)r_T^2$$
$$cov_{T+1} = \lambda cov_T + (1-\lambda)r_{X,T}r_{Y,T}$$

  So, the covariance forecast for day T is $0.15 \times 0.0125 \times 0.0125 = 2.34 \times 10^{-5}$.

  We need to update the variance estimates for X & Y & cov(X,Y):

  $\text{Variance}(X) = 0.92 \times (0.0125^2) + (0.08) \times (0.005^2) = 0.00014575$, $\text{Std}(X) = 1.21\%$.

  $\text{Variance}(Y) = 0.92 \times (0.0125^2) + (0.08) \times (-0.005^2) = 0.00014575$, $\text{Std}(Y) = 1.21\%$.

  $\text{Cov}(X,Y) = 0.92 \times (2.34 \times 10^{-5}) + 0.08 \times (0.005) \times (-0.005) = 1.956 \times 10^{-5}$.

  $\text{Corr}(X,Y) = \text{Cov}(X,Y)/(\text{Std}(X) \times \text{Std}(Y)) = 13.4\%$.

C 选项错误。

- 15.0% is the incumbent correlation

D 选项正确。

- 16.2% is the value if use 0.5% instead of 0.05% for the return on Y

考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]
科目: Financial Markets and Products

5. According to GARP, each of the following was a causal factor in the 2007-2009 global financial crisis (GFC) EXCEPT which is not a causal factor?

A. Low interest rates

B. The originate-to-distribute (OTD) business model and securitization, especially CDOs

C. An unexpected spike in prepayments due to an acceleration in repeat refinancing

D. Dubious lending practices and risky mortgage loan products (e.g., NINJA) and loan features (e.g., teaser rates)

答案:C
解析:
A 选项错误。

- Low interest rates (note the dual role in terms of housing demand and investor supply): "Growth in housing demand and concomitant mortgage financing was fueled (in part) by the low-interest-rate environment that existed in the early 2000s. This demand helped drive substantial increases in housing prices. Low interest rates also spurred investors, including institutional investors, to look for investments that offered yield enhancement. They found this yield in subprime mortgages, which typically carry premiums of up to 300-basis points over the rates charged to prime borrowers."—Source: 2020 FRM Part I: Foundations of Risk Management, 10th Edition

B 选项错误。

- The shift toward banks'originate-to-distribute (OTD) business model which enabled a moral hazard by discouraging careful scrutiny of borrower creditworthiness.

C 选项正确。

- Refinancing enabled borrowers to avoid higher interest rates (as the reset rate) after their initial, teaser rates; the PROBLEM was that home price declines halted the refinance practice that fueled much of the subprime borrowing. In this way, prepayments were not really a cause. In regard to (A), (B) and (D), each is TRUE. Although there were several causes (some disputed), according to GARP's Chapter 10, causal factors included:

D 选项错误。

- Risky mortgage loan products, in particular, teaser rates; e.g., 2/28 adjustable-rate mortgages. (But also interest-only teaser periods and other "innovations"like NINJA loans).

考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]
科目: Financial Markets and Products

6. A hedge fund's big data algorithm can predict the market's direction on five out of eight days (62.5%). Each day's prediction is either a success (e.g., market goes up and algo predicts up) or a failure (e.g., market goes down but algo predicts up). If we dubiously assume the predictions are independent, the binomial distribution fits a series of daily predictions over, say, a week or a month. Over two months, the probability of each day's predication being successful, p, equals 5/8 or 62.5% and the number of days, n, equals 60. We observe that n*p = 60*62.5% = 37.5 and n*(1-p) = 60*37.5% = 22.5, and both of these values (i.e., 37.5 and 22.5) are greater than 10; this satisfies a conventional test that says we can use the normal to approximate the binomial. For example, if p were only 1.0%, then n*p = 6, but 6 is less than 10, and such a binomial is deemed to be too skewed to be approximated by the normal distribution. But ours passes the test so we will approximate with the normal distribution. If we do rely on the normal distribution to approximate this binomial where p = 5/8 and n = 60, what is the probability that the algo makes a correct prediction on only half the days or worse; i.e., where X is the number of successful predictions and we approximate with the normal distribution, what is the Pr(X <= 30)?

A. 2.2750%

B. 8.3500%

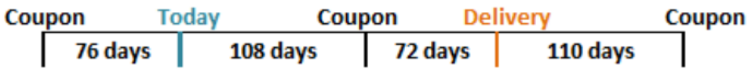
C. 11.090%

D. 14.6667%

答案:A
解析:
A 选项正确。

- The variance $= n \times p \times (1-p)$ and the standard deviation $= \sqrt{n \times p \times (1-p)}$; in this case, the standard deviation $= \sqrt{60 \times (5/8) \times (3/8)} = 3.750$. Where $Z = (30 - 60 \times 5/8)/3.750 = -2.0$, the $\Pr(Z < -2.0) = 2.2750\%$.

7. Suppose that it is known that a bond will be delivered in 180 days under the terms of a futures contract. The timing of coupon payments is shown below. The last coupon on the bond was paid 76 days ago, and the next coupon will be paid in 108 days. The coupon after the next one will be paid in 290 days (i.e., 110 days after delivery). Here is the timeline: next one will be paid in 290 days (i.e., 110 days after delivery). Here is the timeline:



Assume that the risk-free interest rate for all maturities is 5.0% with continuous compounding and that the delivered bond pays a coupon of 9.0% semi-annually with a current quoted (clean) price of USD 124.80 and a conversion factor of 1.300. Which of the following is nearest to the quoted futures price?

A. \$95.00

B. \$98.22

C. \$123.50

D. \$125.27

答案:A
解析:
A 选项正确。

Assumptions	
Face	$100.00
Current Quoted (Clean) Price	$124.80
Coupon	9.0%
Interest rate	5.0%
Conversion Factor	1.300
Delivery (days)	180
Last Coupon (-days)	76
Next Coupon (+ days)	108
Coupon thereafter	182

Timeline	
7/1/2023	Coupon payment
-76 days	
9/15/2023	Current time
108 days	
1/1/2024	(Next) coupon payment
72 days	
3/13/2024	Maturity of futures
110 days	
7/1/2024	Coupon payment

Solution (Hull's Example 6-2)	
Accrued Interest	$1.859
Cash (Dirty, Full) Price	$126.6587
PV of next coupon	$4.434
Cash Futures Price	$125.2760
Days Accrued @ delivery	72
Days Remain @ delivery	110
Futures price, Quoted, 12% bond	$123.4958
Futures price, Quoted, CTD	$94.9968

COC: $F_0=(S_0-I)*EXP(rT)$	
1. COC to calculate Cash forward	
Spot:	$126.659
Income:	$4.434
Forward cash price:	$125.276
2. Then dirty --> clean & standardize	
Subtract AI	$1.780
Quoted Price, 12% bond =	$123.496
Divide by CF of 1.30	$94.997

8. A risk analyst is using the EWMA model to build a daily update of correlation and covariance rates. The two variables are random and are called X and Y. The most recent covariance weight (aka, lambda in the EWMA model) from day n-1 is 0.7. The correlation between X and Y on day N-1 is estimated to be 0.55. X and Y had estimated standard deviations of 0.015 and 0.017 with percentage changes of 3% and 4% respectively on day N-1. The updated covariance between X and Y on day n is nearest to which of the following?

A. 1.48×10^{-4}

B. 4.58×10^{-4}

C. 1.18437

D. 0.04581

答案:B
解析:
B 选项正确。

- We can find the EWMA using the following formula:

  $Cov_N = \lambda \times Cov_{N-1} + (1-\lambda) \times X_{n-1} \times Y_{n-1}$

  $\lambda =$ weight of the most recent covariance on day $n-1$

  $X =$ percentage change for variable X on day $n-1$

  $Y =$ percentage change for variable Y on day $n-1$

  We know that $Cov_{xy} = \rho_{xy} \times \sigma_x \times \sigma_y$

  $Cov(n-1) = 0.55 \times 0.015 \times 0.017 = 0.00014025$

  $Cov_N = 0.7 \times 0.00014025 + (1-0.7) \times 3\% \times 4\% = 0.000458175 = 4.58 \times 10^{-4}$

9. Which of the following is TRUE about the SWIFT (Society for Worldwide Interbank Financial Telecommunication) case study?

A. The 2015-16 cyber-attack (aka, hack) demonstrated that the SWIFT network was unreliable, and it was subsequently phased out

B. The 2015-16 cyber-attack (aka, hack) successfully exploited vulnerabilities to achieve the theft of about USD 81.0 million

C. The 2015-16 cyber-attack (aka, hack) was an unsuccessful attempt to steal money, and it demonstrated the SWIFT network is essentially impervious to attacks

D. The 2015-16 cyber-attack (aka, hack) was a fictitious news account but the negative press nonetheless shook confidence sufficiently in the network that transactions ground to a halt for several weeks

答案:B
解析:
B 选项正确。

- The 2015-16 cyber attack (aka, hack) successfully exploited vulnerabilities to achieve the theft of about USD 81.0 million. In regard to (A), (B), and (D), each is FALSE.

10. A credit portfolio contains some number of independent credit-sensitive assets with identical default probabilities; as the defaults are i.i.d., we can use the binomial distribution to characterize the number of defaults. We are told the expected number of defaults is 4.0 with a variance of 3.80. Which is nearest to the probability of exactly four defaults; Pr(X = 4 | binomial with mean of 4.0 and variance of 3.80)?

A. Less than 0.01%

B. 10.0%

C. 20.0%

D. 33.3%

答案:C
解析:
C 选项正确。

- The mean of a binomial equals $p \times n$ and the variance equals $p \times (1-p) \times n$. We are told that $p \times n = 4.0$ and $p \times (1-p) \times n = 3.80$. We can substitute the first into the second, and thereby determine the number of credits in the portfolio and the default probability: If $p \times (1-p) \times n = [p \times n] \times (1-p) = 3.80$, and $p \times n = 4.0$, then $[p \times n] \times (1-p) = 4.0 \times (1-p) = 3.80$, and $p = 1 - \frac{3.80}{4.0} = 5.0\%$. Therefore, since $n \times p = 4.0$, $n = \frac{4}{0.050} = 80$. The $\Pr(X = 4) = \text{BINOM.DIST}(4, 80 \text{ trials}, 0.050, \text{false}) = 20.043\%$.

11. At the request of his boss at a large commercial bank, Jason is using the bank's internal model to perform a valuation of a freshly constructed pass-through mortgage-backed security (MBS) pool. The pool contains about 500 mortgages with a principal value of $400 billion; all of the mortgages originated within the last 30 days. The pool's weighted average coupon (WAC) is 7.50%. Jason is going to assume a PSA rate of 140%. This multiplies by 1.40 the baseline 100% PSA rate that starts at 0.20% CPR each month for 30 months, then levels at 6.0% CPR. Each of the following is true EXCEPT which is false?

A. The pass-through security will pay investors a coupon rate less than 7.50%

B. The pool factor declines over time due to both scheduled and unscheduled prepayments

C. An increase in the assumption for the PSA rate (e.g., 140% to 170%) implies an increase in present value of this pool

D. The single month mortality (SMM) rate excludes scheduled prepayments, and after three years, it will be about 0.7285%

答案:C
解析:
C 选项错误。
Instead, an increase in the PSA rate models an increase in unscheduled prepayments, which DECREASES the value of the MBS pool.
In regard to (A), (B), and (D), each is TRUE. Specifically,

- The pass-through security will pay investors a coupon rate less than 7.50%
- The pool factor declines over time due to both scheduled and unscheduled prepayments
- The single month mortality (SMM) rate excludes scheduled prepayments, and after three years, it will be about 0.7285%

12. Scott, a risk manager at a small deep-value strategy, is trying to ensure his company is using a coherent risk benchmark. The deep value strategy can be volatile, and thus, Scott wants to see how his capital at risk would adjust if the portfolio had the same holdings but in a smaller proportion to increase its allocation to the risk-free asset. Which principal of a coherent benchmark is reflected in the below scenario?

Portfolio	**Scenario 1**	**Scenario 2**
Portfolio	$20,000,000	$20,000,000
Risk-free Allocation	1.0%	11.0%
Risky Allocation	$19,800,000	$17,800,000
Volatility	24.49%	24.49%
Capital at Risk	$4,850,000	$4,360,101
Risk free Capital	$200,000	$2,200,000
Net Capital at Risk	$4,650,000	$2,160,101

A. Translation Invariance

B. Homogeneity

C. Subadditivity

D. Monotonicity

答案:A
解析:
A 选项正确。

- translation invariance says the addition of (n) reduces the additional cash required by (n). Put more simply, adding cash reduces the risk. For the above example, the increase in the risk-free allocation reduced the risky allocation. The capital at risk was then reduced by the risk-free allocation.

B 选项错误。

- homogeneity refers to the property that multiplying the portfolio size by a constant results in a proportional change in the risk measure.

C 选项错误

- subadditivity states that the sum of two or more portfolios is less than or equal to the sum of their individual risk measures. Adding more assets or asset classes (like commodities) to the portfolio should result in an overall risk measure that is less than or equal to the sum of the individual risk measures.

D 选项错误

- monotonicity implies that if one portfolio has a larger or equal risk than another, then it should also have a larger or equal risk measure. It ensures that adding more assets or increasing allocations to existing assets doesn't increase the risk measure.

13. Barbara is a certified FRM who previously generated income statement and profit projections over a five-year horizon in response to her client's request. Subsequent to the coronavirus outbreak, her client asks Barbara to generate revised financial projections (income statement and balance sheet) and provide the best single point estimate of future revenue, profit, and equity. Barbara utilizes Monte Carlo simulation. Although her client has asked for a single point estimate of these future financial metrics, Barbara perceives that the virus (and the consequential responses) render economic predictions extremely difficult and necessarily laced with great uncertainty. Which of the following is probably her BEST approach to the problem?

A. She can comply by simply selecting the most probable future outcome; the Code has generally nothing to say about this technical task

B. She can comply but she should qualify her findings to avoid overstating the accuracy of her projections; for example, she can plot a future distribution

C. She should work hard to generate the most precise estimate possible; for example, "sales will go down by 10.85%" is more useful than "sales will go down by 10%"

D. She should avoid the temptation to offer any revised future estimates in the face of several "unknown unknowns" because some future values cannot be estimated with sufficient accuracy

答案:B
解析:
B 选项正确。 She can comply, but she should qualify her findings to avoid overstating the accuracy of her projections; for example, she can plot a future distribution.
According to (4.) Fundamental Responsibilities, GARP Members:

4.1 Shall comply with all applicable laws, rules, and regulations (including this Code) governing the GARP Members' professional activities and shall not knowingly participate or assist in any violation of such laws, rules, or regulations.

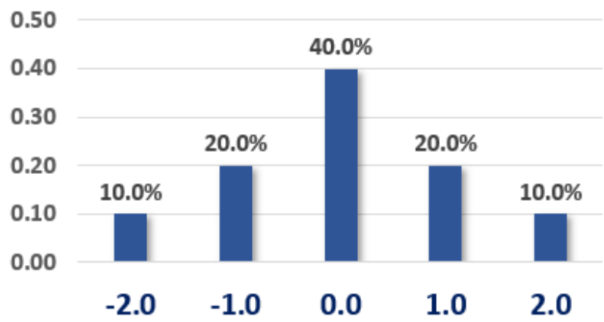
4.2 Shall have ethical responsibilities and cannot out-source or delegate those responsibilities to others.

4.3 Shall understand the needs and complexity of their employer or client and should provide appropriate and suitable risk management services and advice.

4.4 Shall be diligent about not overstating the accuracy or certainty of results or conclusions.

4.5 Shall clearly disclose the relevant limits of their specific knowledge and expertise concerning risk assessment, industry practices, and applicable laws and regulations.[1]

14. Consider the probability mass function (pmf) below. For example, Pr(X = -1.0) = 20.0%.



As we can see, this distribution is symmetrical, so we know that its skewness is zero without performing any calculations. We are told the variance is 1.20 (although we can calculate the variance). What is this distribution's kurtosis?

A. Zero

B. 2.50

C. 3.60

D. 4.40

答案:B
解析:
B 选项正确。
$(-2)^4 \times 10.0\% + (-1)^4 \times 20.0\% + 0^4 \times 40.0\% + 1^4 \times 20.0\% + 2^4 \times 10.0\% = 3.60$, but we need to standardize such that kurtosis $= 3.60/1.20^2 = 2.50$.
Kurtosis is the fourth central moment (aka, fourth moment about the mean) standardized by the standard deviation raised to the fourth power (aka, square of the second moment). So, in the numerator we have $E[X-\mu(x)]^4$, which is the fourth central moment; and this is divided by $\sigma(X)^4$ in order to standardize the moment. See `https://en.wikipedia.org/wiki/Standardized_moment`

15. A U.S. corporate bond that matures on April 1st, 2026, pays a semi-annual coupon of 9.0% per year. The coupon payment dates are Jan 1st and July 1st. Settlement is on April 7th, 2024. The year 2024 is a leap year. The bond's yield happens to be about 7.00%. If the cash (aka, dirty) price is \$1,064.50, then which is nearest to the quoted (aka, clean) price? (Inspired by GARP EOC PQ 17.13)

A. \$960.14

B. \$974.50

C. \$1,040.50

D. \$1,088.50

答案:C
解析:
C 选项正确。 True: \$1,040.50
On a 30/360 basis, there are $30 \times 3 + 6 = 96$ days. The accrued interest (AI) is $\$1,000 \times 9.0\%/2 \times 96/180 = \$24.00$.

Therefore, the bond's quoted price is $\$1,064.50 - 24.00 = \$1,040.50$.
Please note we could have retrieved the cash price (which is given) by starting with the bond price at the last coupon date; \$1,045.151 is the price of the bond on 1/1/2024 where $N = 5$, $I/Y = 3.5\%$, $PMT = 45$, and $FV = 1,000$. Then compounded forward: $\$1,045.151 \times (1 + 7.0\%/2)^{96/180} = \$1064.50$.

16. When is the soonest that an obligation rated AAA (the highest rating) can default?

	AAA	AA	A	BBB	BB	B	CCC to C	Default
AAA	91.00	9.00	-	-	-	-	-	-
AA	1.00	84.00	7.00	5.00	3.00	-	-	-
A	0.40	2.00	91.60	3.00	2.00	1.00	-	-
BBB	-	0.60	2.00	88.50	5.00	3.00	0.90	-
BB	-	0.40	1.00	3.00	79.60	10.00	4.00	2.00
B	-	-	0.80	3.00	9.00	67.20	9.00	11.00
CCC - C	-	-	-	2.00	10.00	15.00	58.00	15.00

When is the soonest that an obligation rated AAA (the highest rating) can default?

A. During the first year (i.e., by the end of the first year) with a margin probability of 2.0%

B. During the second year (i.e., between the end of the first and second year) with a joint probability of 0.1879% (or 0.001879 or 1.879×10^{-3})

C. During the third year (i.e., between the end of the second and third year) with an unconditional probability of 0.00540% (or 0.0000540 or 5.40×10^{-5})

D. During the fourth year (i.e., between the end of the third and fourth year) with a conditional probability of 2.00%

答案:C
解析:
C 选项正确。TRUE: During the third year (i.e., between the end of the second and third year) with an unconditional probability of 0.00540% (or 0.0000540 or 5.40×10^{-5})
At the end of the first year, the worst downgrade (from AAA) is only to AA with a 9.0% probability. Conditional on the AA-rating at the start of the second year, the worst downgrade possible is to BB with a conditional probability of 3.0% (if downgraded to BBB or higher, this obligation cannot default by the end of the third year!). Conditional on the BB-rating at the start of the third year, the obligation can default during the third year with a conditional probability of 2.0%. The probability is given by $9.0\% \times 3.0\% \times 2.0\% = 0.005400\%$.

17. Arguably the U.S. savings and loan (S&L) crisis in the 1980s had multiple causes, but among the following which is the BEST summary explanation of the S&L crisis?

A. S&Ls engaged in a practice called "riding the yield curve" but this was one of the central bank's monetary policy tools and was not meant to be performed by private banks

B. The Fed's restrictive monetary policy led to a dramatic increase in short-term interest rates which wiped out bank's interest rate spread due to their asset-liability mismatch

C. The Fed's expansionary monetary policy led to a dramatic decrease in short-term interest rates which wiped out bank's interest rate spread due to their asset-liability mismatch

D. New regulations introduced in the early 1980s hindered the ability of the savings and loan (S&L) industry to lend and/offer new loan products that could help it grow out of its problems

答案:B
解析:
B 选项正确。 True: The Fed's restrictive monetary policy led to a dramatic increase in short-term interest rates which wiped out bank's interest rate spread due to their asset-liability mismatch.

As explained by GARP, ”However, rising inflation in the late 1970s prompted the Fed to implement a restrictive monetary policy, which led to a significant increase in short-term interest rates. The increase in short-term rates pushed up funding costs for S&Ls, wiping out the interest rate spread they depended on for their profit margin. The spike in their short-term funding costs (which were needed to finance long-term fixed-interest rate mortgages) meant that S&Ls generated negative net interest margins on many of their long-term residential mortgage portfolios.”[1]

[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions, 10/2019

18. A discrete random variable is characterized by the probability mass function (pmf) as given by $f(x) = x \cdot a$, and its domain is the set of integers $\{6, 7, 8, 9, 10\}$. What is the variable's expected value?

A. 6.67

B. 8.00

C. 8.25

D. 9.33

答案:C

解析:

C 选项正确。 True: 8.25

We know that $6a + 7a + 8a + 9a + 10a = 1.0$ such that $a = 1/40 = 0.0250$ and the pmf is given by $f(x) = x/40 = 0.0250 \times x$. The expected value equals $1/40 \times (6^2 + 7^2 + 8^2 + 9^2 + 10^2) = 8.25$.

19. Assume the following six-month forward rates that are quoted per annum with semi-annual compounding:

- $F(0, 0.5) = S(0.5) = 2.0\%$; i.e., the six-month spot rate
- $F(0.5, 1.0) = 4.0\%$
- $F(1.0, 1.5) = 6.0\%$

If the price of a 7.0% semi-annual coupon bond is $103.20, which is nearest to the 2-year spot rate?

A. 4.30%

B. 5.40%

C. 6.60%

D. 7.90%

答案:B

解析:

B 选项正确。 True: 5.40%

20. In regard to through-the-cycle (TTC) versus at-the-point (aka, point in time, PIT) approaches to credit ratings, each of the following statements is true EXCEPT which is false?

A. Agency (i.e., external) credit ratings tend to be through-the-cycle (TTC)

B. Through-the-cycle (TTC) is conditional, while at-the-point (PIT) is unconditional

C. Credit spreads are at-the-point (PIT) and PIT measures incorporate more information

D. During crisis periods, PIT approaches imply higher expected and unexpected loss (EL & UL) such that PIT tends to be pro-cyclical

答案:B
解析:
B 选项错误。 Instead, true is: Through-the-cycle (TTC) is unconditional, while at-the-point (PIT) is conditional.
In regard to (A), (C) and (D), each is TRUE.
In regard to true (A), external agency credit ratings tend to be through-the-cycle (TTC).
In regard to true (C), credit spreads are at-the-point (PIT) and PIT measures theoretically incorporate more information. Please note this point is debatable. Some people would argue that much of the information in some PIT approaches (e.g., structural) contain too much noise masquerading as information.
In regard to true (D), during crisis periods, PIT approaches imply higher expected and unexpected loss (EL & UL) such that PIT tends to be pro-cyclical.

21. Alice, Bert, Chris, Don, Eva, and Fred are individual investors. Each of them exhibits a particular behavioral bias.

- Alice (A) invested in the stock Cloudera (CLDR) and bad news (aka, new information) renders her original thesis obsolete but she is reluctant to sell today because an exit implies the realization of a -30.0% loss on the position and she much prefers to sell after her position experiences a double-digit gain
- Bert (B) can buy a new smartphone for $79.00 but he cannot resist a sale and prefers to pay $100.00 because it represents a 50.0% discount from the retail (MSRP) price
- Chris (C) attends an investment conference but he avoids the seminar focusing on recession risks because he is overweight homebuilders and he worries the topic will make him anxious with worry
- Don (D), who enjoys food shopping, tends to be price-conscious (e.g., he seeks bargains) when he pays cash at Sprouts or Trader Joes, but when he uses his Amazon credit card at Whole Foods he doesn't worry about the cost because he doesn't look at the statement for several days or weeks
- Eva (E) purchased Facebook (FB) at $160.00 because his firm's price target was $200.00, and he decides to ignore new information until the price reaches this level
- Fred (F) was previously a patient buy-and-hold investor who purchased high-conviction stocks and only checked his portfolio once a week, but last year he signed up for a subscription to Seeking Alpha and since that time his buy/sell transactions have quintupled because he's reading news about his portfolio holdings every day

Which of the following correctly matches the individual investor to his or her behavioral bias?

A. A = Home bias, B = Mental accounting, C = Ostrich effect, D = Herding, E = Groupthink, F = Framing

B. A = Loss aversion, B = Framing, C = Ostrich effect, D = Mental accounting, E = Anchoring, F = Feedback effects

C. A = Groupthink, B = Herding, C = Home bias, D = Framing, E = Loss aversion, F = Ostrich effect

D. A = Mental accounting, B = Home bias, C = Loss aversion, D = Groupthink, E = Feedback effects, F = Herding

答案:B
解析:
B 选项正确。 TRUE: A = Loss aversion, B = Framing, C = Ostrich effect, D = Mental accounting, E = Anchoring, F = Feedback effects
According to GARP's Box 8.3:[1]

- **Anchoring and referencing:** This is the use of mental reference points to contextualize a decision (e.g., such as using an existing price point to determine whether a new price point is attractive or not). The anchor may influence the decision-making process in an irrational way. Furthermore, the various reference points in a collection of related decisions may lack coherence.
- **Feedback effects:** The presence or absence of frequent, positive feedback can irrationally influence the ability of decision-makers to stick to a decision.

- **Framing:** How a choice is framed can push a decision-maker toward one decision or another. For example, a consumer may be willing to hunt for a 50% savings on a phone case (saving themselves USD 10) but be unwilling to make the same effort to save the USD 10 when buying a USD 200 phone (because it represents a smaller percentage of the purchase price).

- **Groupthink:** This describes the tendency of individuals within groups to overcome their doubts about a risky decision (or keep quiet) in favor of the group consensus. The consensus may itself have been shaped by a dominant individual, poorly set targets, or selective reading of ambiguous evidence.

- **Herding:** Herding is the tendency of investors to copy the actions of others, both when investing and when reducing losses in a volatile market. Herding effects in risk management can lead to too many investors using the same risk metrics or setting the same stop-losses, leading to sharp market sell-offs.

- **Home bias:** This describes the tendency of investors to invest in domestic securities rather than building a globally diversified portfolio, perhaps because of the uncertainties attached to foreign markets.

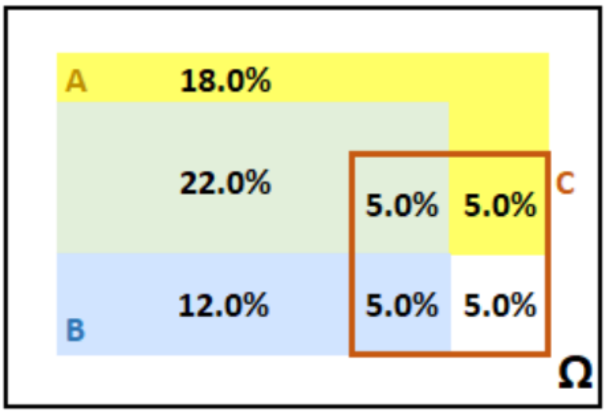
- Loss aversion: Experiments show that for most people the potential for losses outweighs potential gains of similar magnitude. This can lead a decision-maker to favor a result that is presented as certain while foregoing the chance of larger but riskier wins. Loss aversion does not always lead to conservative risk decisions. It can also encourage decisionmakers to take irrationally risky decisions to preserve some chance of avoiding a loss. (Whether a decision is framed as a loss or as a potential gain is also therefore important.)

- Mental accounting: People seem to account for money within separate categories that are treated differently as if the money was not completely fungible across accounts. For example, consumers might spend more if they use a credit card compared to using cash. Investors might invest the money from an inheritance differently than money from a gambling win. They may also be reluctant to "close"a mental account if it involves declaring a loss or mistake. Loss aversion and other behavioral phenomena, such as the treatment of "sunk"costs, often further distort mental accounting.

- Ostrich effect: This describes the irrational tendency to avoid observing bad news that might precipitate uncomfortable decisions or actions. For example, an investor might pay more attention to booming stock markets than flat or falling markets. (Conversely, an investor that pays too much attention to each individual loss can suffer from irrational aversion.)

[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions, 10/2019

22. The probability graph below illustrates event A (the yellow rectangle) and event B (the blue rectangle). The unconditional probability of event A is 50.0% and the unconditional probability of event B is 44.0%; i.e., $\Pr(A) = 50.0\%$ and $\Pr(B) = 44.0\%$. Their overlap is graphed by the green rectangle such that $\Pr(A \cap B) = 27.0\%$. The orange rectangle conditions on the event C. For example, conditional on event C, there is a 50.0% probability that event A occurs, $\Pr(A|C) = 50.0\%$.



Which of the following is TRUE about, respectively, the unconditional and conditional relationship between events A and B?

A. A and B are unconditionally dependent but conditionally (on event C) independent

B. A and B are unconditionally dependent and also conditionally (on event C) dependent

C. A and B are unconditionally independent but conditionally (on event C) dependent

D. A and B are unconditionally independent and also conditionally (on event C) independent

答案:A
解析:
A 选项正确。 True: A and B are unconditionally dependent but conditionally (on event C) independent.
If A and B are unconditionally independent, then it must be true that $\Pr(A \cap B) = P(A) \times P(B)$; however, in this case, $P(A) \times P(B) = 50.0\% \times 44.0\% = 22.0\%$, but $\Pr(A \cap B) = 27.0\%$. Therefore, A and B are unconditionally dependent.
On the other hand, A and B are conditionally independent because it is true that $\Pr(A \cap B|C) = \Pr(A|C) \times \Pr(B|C)$. The $\Pr(A \cap B|C) = 5.0\%/20.0\% = 25.0\%$, and $\Pr(A|C) \times \Pr(B|C) = (10.0\%/20.0\%) \times (10.0\%/20.0\%) = 0.50 \times 0.50 = 25.0\%$.

23. We are told to assume that an interest rate is 9.00% per annum with quarterly compounding. Each of the following statements is true EXCEPT which is false?

A. The effective annual rate is higher than 9.00%

B. The equivalent rate with continuous compounding is higher than 9.00%

C. The equivalent rate with monthly compounding (i.e., 12 periods per year) is lower than 9.00%

D. The stated (aka, quoted, nominal) rate is given as 9.00% such that the periodic rate is 2.25%

答案:B
解析:
B 选项错误。 Instead, the equivalent rate with continuous compounding is LOWER than 9.00%.
In regard to (A), (C), and (D), each is TRUE.
Specifically, if an interest rate is 9.00% per annum with quarterly compounding, then:

- The equivalent continuous rate is $4 \times \ln(1 + 9\%/4) = 8.90\%$
- The equivalent rate with monthly compounding is $[(1 + 9\%/4)^{4/12} - 1] \times 12 = 8.933\%$
- The effective annual rate is $(1 + 9\%/4)^4 - 1 = 9.3083\%$; i.e., the equivalent rate with annual compounding

If the stated (aka, quoted, nominal) rate is given as 9.00%, and there are four periods per year (aka, quarterly compounding), then the periodic rate is $9.00\%/4 = 2.25\%$.

24. Quotecan Corporation has issued a bond. Quotecan's interest coverage ratio is within reasonable limits, providing adequate protection for investors in the near term. The bond is "better than junk" but carries more credit risk than bonds issued by the highest-rated corporations (e.g., Microsoft, Apple, Exxon Mobil, Johnson & Johnson, and Walmart). Quotecan's capacity to repay may be impaired if adverse economic conditions materialize, or if circumstances suddenly change. While the obligations are not speculative in their entirety, they do contain just a few speculative elements which may render the protective elements unreliable over a longer time horizon. Based on these facts, which credit rating is most likely assigned to Quotecan's bond offering?

A. A-1 or P-1

B. AA or Aa

C. B or B

D. BBB or Baa

答案:D
解析:
D 选项正确。 True: BBB (Standard & Poor's) or Baa (Moody's) rating is most likely; i.e., it is the lowest rating among the investment-grade ratings. See `https://en.wikipedia.org/wiki/Credit_rating`.

According to S&P, an obligation rated ”BBB” exhibits adequate protection parameters. However, adverse economic conditions or changing circumstances are more likely to weaken the obligor's capacity to meet its financial commitments on the obligation. According to Moody's, the rating of ”Baa” corresponds to obligations of moderate credit risk that may possess speculative characteristics.

25. GARP explains that ”perhaps the biggest argument for ERM is that an enterprise-level perspective is the best way to prioritize risks and optimize risk management.” Each of the following is one of the four key reasons that enterprise risk might demand the practice of (or, the art and science of) enterprise risk management, ERM, EXCEPT which is NOT one of the four key reasons?

A. Identify hidden and/or dangerous concentrations including (for example) geographical, industry, product, or among suppliers

B. Vertical vision recognizes potential threats to the whole enterprise ideally in their early stages to reduce the leverage effect of time

C. Portfolio thinking permits the business units to specify a more accurate distribution that incorporates the allocation of corporate overhead

D. Thinking beyond silos acknowledges diversification among risk types and better understanding of how interactions might worsen enterprise threats

答案:C
解析:
C 选项错误。 Instead, the fourth is: translate a better understanding of risks to the enterprise, in particular insurable risks, into costs savings and/or better resource allocation.
The four key reasons why enterprise risk might demand ERM are:

- Identify hidden and/or dangerous concentrations including (for example) geographical, industry, product, or among suppliers
- Vertical vision recognizes potential threats to the whole enterprise ideally in their early stages to reduce the leverage effect of time
- Translate a better understanding of risks to the enterprise, in particular insurable risks, into costs savings and/or better resource allocation
- Thinking beyond silos acknowledges diversification among risk types and better understanding how interactions might worsen enterprise threats

26. Yesterday a web page hosted by Acme received tens of thousands of page views but some were views by malicious bots. Acme utilizes two software applications to detect these malicious ”bot-views.” It uploads the same data file from yesterday to both applications. The first application detects 200 bot-views and the second application detects 300 bot-views. Among these, only 40 bot-views were detected by both applications. All bot-views are equally likely to be located, but clearly both applications only identify a minority of the bot-views (otherwise there would be a much higher number of identified bot-views common to both applications). Further, the identification of a bot-view by one application is independent of its identification by the other application. How many malicious bot-views did the web page experience on this day?

A. 300

B. 460

C. 540

D. 1,500

答案:D
解析:
D 选项正确。 True: 1,500
The second application identified $40/200 = 20.0\%$ of those identified by the first application, therefore (per the independence), we can infer that its own 300 identifications is about 20.0% of the total number such that we estimate $300/20\% = 1,500$ total bot-views. Similarly, the first application identified $40/300 = 13.33\%$ of those identified by the second application, so we can infer that its own 200 identification represents about 13.33% of the total, which is also $200/13.33\% = 1,500$.

27. Consider a binary cash-or-nothing call option that pays \$100.00 if the stock is above \$100.00 at maturity in three months. The option's underlying non-dividend-paying stock is currently trading at \$85.00. The riskfree rate is 4.0% per annum with continuous compounding. We are also told that while $S = \$85.00$ and $K = \$100.00$, $N(d_1) = 0.12350$ and $N(d_2) = 0.100$, and the price of a vanilla three-month European call option on this stock is \$0.62. Which is nearest to the price of the cash-or-nothing call?

A. \$0.62

B. \$9.88

C. \$10.50

D. \$20.38

答案:B
解析:
B 选项正确。 True: \$9.88
$\$100.00 \times 0.10 \times \exp(-4.0\% \times 0.25) = \$9.88$.
Alternatively, we can retrieve the price of the asset-or-nothing as given by $\$85.00 \times 0.12350 \times \exp(0\% \times 0.25) = \$10.50$ such that $\$10.50 - \$0.62 = \$9.88$.

28. Rebecca is evaluating four different countries in an attempt to determine their respective default risk. She evaluates the countries in four categories: degree of indebtedness as measured by debt as a percentage of gross domestic product; social service/pension commitments as estimated by the average age of the population; nature of the economy (e.g., diverse versus concentrated in oil as a natural resource), and monetary policy. The four countries are summarized in the exhibit below:

Country	**Debt as % of GDP**	**Age of Population**	**Economy**	**Monetary Policy**
Country A	$> 100\%$	Old	Diverse	Independent CB
Country B	$\sim 100\%$	Young	Tourism	EU member
Country C	50%	Young	Oil	Autocracy
Country D	30%	Old	Diverse	Semi-independent CB

Which of the four countries is most likely to default?

A. Country A is the most likely to default because debt above 100% is highly predictive of default and overwhelms the other indicators

B. Country B is the most likely to default because it is the only country with three (out of four) negative indicators

C. Country C is the most likely to default because it has an autocratic government

D. It is unclear: each has two negative (and two positive) indicators; further, quantification of this scorecard is an insufficient predictor

答案:D
解析:

It is unclear: each has two negative (and two positive) indicators; further, quantification of this scorecard is an insufficient predictor.
Damodaran: “In summary, a full assessment of default risk in a sovereign entity requires the assessor to go beyond the numbers and understand how the country’s economy works, the strength of its tax system and the trustworthiness of its governing institutions.”[1]
In regard to (A), (B) and (C), each is not the best answer:
In regard to (A), degree of indebtedness is “the most logical place to start assessing default risk” but “[the list of 20 countries that owe the most] suggests that this statistic (government debt as percent of GDP) is an incomplete measure of default risk. The list includes some countries with high default risk (Greece, Congo, Jamaica, Egypt) but it also includes some countries that were viewed as among the most creditworthy by ratings agencies and markets (US, Japan, France and Canada).”[1]
In regard to (B), Country B has only two negative indicators: high debt/GDP and an economy concentrated in tourism. But in regard to Damodaran casts EU membership as net positive: “6. Implicit backing from other entities: When Greece, Portugal and Spain entered the European Union, investors, analysts and ratings agencies reduced their assessments of default risk in these countries. Implicitly, they were assuming that the stronger European Union countries – Germany, France and the Scandinavian countries – would step in to protect the weaker countries from defaulting. The danger, of course, is that the backing is implicit and not explicit, and lenders may very well find themselves disappointed by lack of backing, and no legal recourse.”[1]
In regard to (C), Damodaran: “5. Political risk: Ultimately, the decision to default is as much a political decision as it is an economic decision. Given that sovereign default often exposes the political leadership to pressure, it is entirely possible that autocracies (where there is less worry about political backlash) are more likely to default than democracies. Since the alternative to default is printing more money, the independence and power of the central bank will also affect assessments of default risk.”[1]

29. GARP explains that the Basel Committee on Banking Supervision’s standard number 239 (aka, BCBS 239) ”was a major driver in the rise of the chief data officer (CDO) function. The CDO is typically responsible for standardizing a firm’s approach to data management.”[1] In addition, which of the following statements is also TRUE about the Basel Committee on Banking Supervision’s standard number 239 (aka, BCBS 239)
[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions, 10/2019

A. BCBS 239 offers the advantage of a uniform blueprint for a compliant IT infrastructure

B. Every systematically important financial institution (SIFI) complied with the original timeline

C. Compliance with BCBS 239 is a subjective exercise and the standards for each bank are bespoke

D. While praised for its theory, critics say BCBS 239 forgets to address risk management data and models

答案:C
解析:
C 选项正确。 TRUE: Compliance with BCBS 239 is a subjective exercise and the standards for each bank are bespoke.
In regard to (A), (B) and (D) each is FALSE. Instead, the following are TRUE statements:

- ”There is no uniform blueprint in place for a BCBS 239-compliant infrastructure and solutions are specific to each institution. The optimal approach ensures that all people and systems within the banking group are working with the same data, the same models, and the same assumptions,” explains GARP.[1]

- ”Banks have struggled to comply with BCBS 239 and the original timeline to achieve full compliance was not met by any bank. This is largely due to the highly complex nature of the IT reengineering involved in bringing the various systems into compliance as well as the dynamic nature of the principles. The exponential increase in the application of AI techniques on large data sets has also made compliance with BCBS 239 more challenging,” explains GARP.[1]

- While implementation has lagged, BCBS 239 does (at least attempt to) address risk management data and models in a practical way.

[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions, 10/2019

30. For her investment firm, Audrey has identified six high-priority machine learning use cases. They are given below in three pairs:

- I. Segment customers into different groups; and perform sentiment analysis of 10K SEC filings
- II. Predict if borrower(s) will default; and forecast future stock price(s) based on fundamental data
- III. Decide how to split large-volume trades to sell quickly while minimizing the adverse price effect, and decide how much of a position to hedge using derivatives.

The three top-level machine learning approaches are supervised learning (S), unsupervised learning (U), and reinforcement learning (R). Which of the following is the best mapping of the advisable machine learning approach to its corresponding pair of use cases above?

A. SRU: I=Supervised, II=Reinforcement, III=Unsupervised

B. RUS: I=Reinforcement, II=Unsupervised, III=Supervised

C. RUU: I=Reinforcement, II=Unsupervised, III=Unsupervised

D. USR: I=Unsupervised, II=Supervised, III=Reinforcement

答案:D
解析:
D 选项正确。 True: USR: I=Unsupervised, II=Supervised, III=Reinforcement
These pairs map as follows:

- I. Unsupervised: Customer segmentation implies clustering, and sentiment analysis implies natural language processing (NLP).
- II. Supervised: Default prediction and asset price forecasting are classic applications of supervised learning.
- III. Reinforcement: As GARP explains, "Reinforcement learning has many applications in risk management. For example, it is used to determine the optimal way to buy or sell a large block of shares, to determine how a portfolio should be managed, and to hedge derivatives portfolios [section 14.1] ...As mentioned earlier, there are also many potential applications in finance, including for technical trading rules, determining how to split large-volume trades to sell quickly while minimizing the adverse price effect, and determining how much of a position to hedge using derivatives." [Section 14.7]

31. A non-dividend-paying stock has a current price of $10.00 and a volatility of 25.0% per annum. The risk-free rate is 4.0% per annum with continuous compounding. Consider the following four spread trades:

- Trade #1 is a bull spread with calls. The strike prices are: $K_1 = \$9.00$, $K_2 = \$11.00$. The up-front cost is $1.00.
- Trade #2 is a bull spread with puts. The strike prices are: $K_1 = \$9.00$, $K_2 = \$11.00$. The up-front cash flow is $0.93.
- Trade #3 is a bear spread with puts. The strike prices are: $K_1 = \$8.00$, $K_2 = \$12.00$. The up-front cost is $1.84.
- Trade #4 is a bear spread with calls. The strike prices are: $K_1 = \$8.00$, $K_2 = \$12.00$. The up-front cash flow is $2.00.

In regard to each trade's maximum profit, each of the following statements is true EXCEPT which is false?

A. Trade #1 offers a maximum profit of $1.00

B. Trade #2 offers a maximum profit of $0.93

C. Trade #3 offers a maximum profit of $0.84

D. Trade #4 offers a maximum profit of $2.00

答案:C
解析:
C 选项错误。 Instead, Trade #3 offers a maximum profit of \$2.16. In regard to (A), (B), and (D), each is TRUE.

32. Political Risk Services (The PRS Group) provides numerical measures of country risk for more than a hundred countries. Its International Country Risk Guide (ICRG) rating "comprises 22 variables in three subcategories of risk: political, financial, and economic. A separate index is created for each of the subcategories. The Political Risk index is based on 100 points, Financial Risk on 50 points, and Economic Risk on 50 points. The total points from the three indices are divided by two to produce the weights for inclusion in the composite country risk score."[1]
As of July 2018, according to Damodaran, Taiwan's composite (ICRG) score was 85. Which of the following is TRUE?

A. Taiwan is very low risk

B. This score is irrelevant for investment purposes

C. This score can be neither normalized nor standardized

D. This score cannot be used to rank Taiwan against other countries

答案:A
解析:
A 选项正确。 TRUE: Taiwan is very low risk. The composite scores, ranging from zero to 100, are then broken into categories from Very Low Risk (80 to 100 points) to Very High Risk (zero to 49.9 points).
In regard to (B), (C), and (D), each is FALSE.
In regard to false (B), "irrelevant" goes too far for an index that assigns 25% weight (50 of 200 points) to Financial Risk and 25% to Economic Risk. Damodaran's point is about not so severe and directed in general: "Measurement models/methods: Many of the entities that develop the methodology and convert them into scores are not business entities and consider risks that may have little relevance for businesses. In fact, the scores in some of these services are more directed at policymakers and macroeconomists than businesses."[1]
In regard to false (C), to normalize is to translate to $(0,1)$ scale. To standardize is to translate into standard normal terms, $N(0,1)$. We can do either to the PRS scores.
In regard to false (D), the scores are ordinal (can be used for ranking) but neither interval nor ratio. Damodaran: "More rankings than scores: Even if you stay with the numbers from one service, the country risk scores are more useful for ranking the countries than for measuring relative risk. Thus, a country with a risk score of 80, in the PRS scoring mechanism, is safer than a country with a risk score of 40, but it would be dangerous to read the scores to imply that it is twice as safe."[1]
[1] Aswath Damodaran, Country Risk: Determinants, Measures and Implications –The 2018 Edition (July 23, 2018). See also `https://www.prsgroup.com/explore-our-products/international-country-risk-guide/`

33. Consider the following three portfolios where we know the expected excess returns of Portfolios A and B; i.e., $E[ER(A)] = 6.0\%$ and $E[ER(B)] = 8.4\%$ and these in excess of the riskfree rate. For Portfolios A and B, we also know the factor betas: $\beta(A,1) = 0.40$, $\beta(A,2) = 1.20$, $\beta(B,1) = 0.80$, $\beta(B,2) = 1.50$. The unknowns are the two risk premiums on the two factors, $F(1)$ and $F(2)$:

- Portfolio A: $E[ER(A)] = \beta(A,1) \times F(1) + \beta(A,2) \times F(2) = 0.40 \times F(1) + 1.20 \times F(2) = 6.0\%$
- Portfolio B: $E[ER(B)] = \beta(B,1) \times F(1) + \beta(B,2) \times F(2) = 0.80 \times F(1) + 1.50 \times F(2) = 8.4\%$
- Portfolio C has the following betas: $\beta(C,1) = 1.30$ and $\beta(C,2) = 0.90$.

If Portfolio C has betas as given by $\beta(C,1) = 1.30$ and $\beta(C,2) = 0.90$, what is the expected excess return of Portfolio C?

A. 5.00%

B. 7.50%

C. 10.00%

D. Not enough information

答案:B

解析:

B 选项正确。 7.50%. We have two equations and two unknowns such that it must be the case that $F(1) = 3.0\%$ and $F(2) = 4.0\%$. If the betas for Portfolio C are $\beta(C,1) = 1.30$ and $\beta(C,2) = 0.90$, the expected excess return of Portfolio C is given by $E[ER(C)] = 1.30 \times 3.0\% + 0.90 \times 4.0\% = 7.50\%$.

34. Robert is trying to price an exotic path-dependent option (Price(E)) for which no analytical solution exists. He uses a Black-Scholes option pricing model (BSM OPM) for vanilla options, referring to its analytical price as BSM(V). He also employs a Monte Carlo simulation (MCS) to estimate the exotic option's price as MCS(E). He re-uses the same simulated paths to get a simulated value for the vanilla option, MCS(V). He then applies an error correction formula:

$Price(E) = MSC(E) + \beta \times [BSM(V) - MCS(V)]$.

The goal is to improve accuracy by reducing variance. The text states that a good control variate should be inexpensive to construct, and asks for the other property that Bob seeks.

A. High correlation between Price(E) and BSM(V)

B. An analytical function for the price of the path-dependent option, Price(X)

C. A random number generator (RNG) that is neither pseudo-random nor deterministic but genuinely random

D. A second set of random variables that by construction have a negative correlation with the iid variables used in the simulation

答案:A

解析:

A 选项正确。 TRUE: High correlation between Price(E) and BSM(V)

Using the notation of the reading, the unknown distribution is $g(X)$ such that the Monte Carlo simulation approximates the expected value, $E[g(X)]$, but its approximation can be decomposed as $E[g(X(i))] + \eta(i)$ where $\eta(i)$ is a mean zero error. The control variate is another random variable, $h(X(i))$, that is correlated with the error. In this question, Bob is estimating $E[Price(E)] = g(X)$ while BSM(V) is the control variate.

According to the reading, "a good control variate should have two properties:

1. First, it should be inexpensive to construct from x_i. If the control variate is slow to compute, then larger variance reductions—holding the computational cost fixed—may be achieved by increasing the number of simulations b rather than constructing the control variates.

2. Second, a control variate should have a high correlation with $g(X)$. The optimal combination parameter β that minimizes the approximation error is estimated using the regression: $g(x(i)) = \alpha + \beta h(X(i)) + v(i)$.

35. Every year, Acme Corporation grants stock options to its executives and employees. Like most of its peers, the employee stock options (ESOs) vest over four years and always have a 10-year term. In the event of termination, depending on the particulars, options sometimes accelerate and sometimes are forfeited. Newly recruited executives often receive abnormally large, front-loaded option grants. All option grants always have a fixed strike price, and to date, Acme has never re-priced grants. In the majority of cases, Acme grants at-the-money (ATM) options; in rare cases, Acme grants out-of-the-money (OTM) options. However, Acme never grants in-the-money (ITM) ESOs. To value ESOs, the company has developed basic and advanced versions of both the Black-Scholes-Merton (BSM) and the binomial (aka, lattice) models.

Which of the following statements is TRUE?

A. In theory, the binomial model is better than the BSM model to price the company's ESOs

B. The ESO grants do not dilute the company's shareholders because the company never grants in-the-money options

C. The ESO grants do not reduce the company's reported profit because the ESOs have zero intrinsic value at grant

D. The fair market value (FMV) of the ESOs is equal to the standard BSM value assuming a 10-year term; i.e., ESO FMV = $c = c(S, K, \sigma, 10, r, q)$

答案:A

解析:

A 选项正确。 True. In theory, the binomial model is better than the BSM model to price the company's ESOs because they are American-style options (with vesting/forfeiture restrictions that are naturally handled by the binomial) rather than European-style options. Please note this solution only says the binomial is "better": many companies do employ the BSM model for ESOs and (for example) use their shorter expected life (or expected lives) as the maturity input in lieu of the 10 year term.

In regard to (B), (C), and (D), each is false. Instead, the corresponding TRUE statements are:

- ESO grants do dilute the company's shareholders
- ESO grants do reduce the company's reported profit because they are expensed per accounting rules
- ESOs are worth less than the European-style call option value given by $c = c(S, K, \sigma, 10, r, q)$. This is due to (i) vesting restrictions and (ii) their relative illiquidity (in contrast to the liquidity of regular call options)

36. A bond portfolio with a face value of \$7.0 million is exposed to the following two key rates:

- The 2-year key rate, KR01(2-year) is equal to \$0.030 per 100 face amount and has a daily volatility, σ(bps), of 12.0 basis points
- The 5-year key rate, KR01(5-year) is equal to \$0.090 per 100 face amount and has a daily volatility, σ(bps), of 25.0 basis points

As the curve experiences significant non-parallel shifts, the correlation between the key rates is only 0.440. Which of the following is nearest to the daily volatility of the portfolio (in dollars)?

A. \$159,500

B. \$170,100

C. \$182,700

D. \$219,300

答案:B

解析:

B 选项正确。 True: \$170,100.

In dollars, portfolio volatility $= \sqrt{(\$2,100.0)^2 \times (12.0)^2 + (\$6,300.0)^2 \times (25.0)^2 + 2 \times \$2,100.0 \times \$6,300.0 \times 12.0 \times 25.0 \times 0.440} =$ $\$170,000.00$; where $\$2,100 = \$0.030 \times \$7,000,000/100$ and $\$6,300 = \$0.090 \times \$7,000,000/100$.

In percentage terms, portfolio volatility $= \sqrt{(0.00030)^2 \times (12.0)^2 + (0.00090)^2 \times (25.0)^2 + 2 \times 0.00030 \times 0.00090 \times 12.0 \times 25.0 \times 0}$ 2.430%, and $2.430\% \times \$7,000,000 = \$170,100.00$.

37. During a 36-month period during which the risk-free rate was 2.0%, consider a comparison between the market portfolio and two fund managers, Betty and Peter:

- The market portfolio's excess return was +6.0% (its gross return was +8.0%) with a volatility of 24.0% per annum
- Peter's Portfolio: Excess return = +7.0% (gross return = +9.0%), Volatility = 36.0% per annum, Beta $\beta(P, M) = 0.750$
- Betty's Portfolio: Excess return = +11.0% (gross return = +13.0%), Volatility = 44.0% per annum, Beta $\beta(B, M) = 1.650$

If the capital asset pricing model (CAPM) is valid, then each of the following statements is true EXCEPT which is false?

A. If the investor is diversified (or the portfolio represents only a fraction of the investor's entire wealth), the Treynor (TPI) is a good measure and Peter's TPI is better than Betty's

B. If the investor is not diversified (or the portfolio represents the investor's entire wealth), the Sharpe (SPI) is a good measure and Betty's SPI is better than Peter's

C. Betty's portfolio already lies on the efficient frontier such that we do not expect her to sustain further improvement in the reward-to-variability ratio

D. If Betty's and Peter's portfolio belong to different peer groups, Jensen's alpha is a good measure for ranking them and Betty's (Jensen's) alpha is better than Peter's

答案:D

解析:

D 选项错误。 False, doubly: if the portfolios belong in SIMILAR peer groups (i.e., similar risk levels), Jensen's alpha is good for ranking and Peter's is better than Betty's.

In regard to (A), (B), and (C), each is TRUE.

For (A), If the investor is diversified (or the portfolio represents only a fraction of the investor's entire wealth), the Treynor (TPI) is a good measure. Peter's TPI is 9.33% and Betty's TPI is 6.67%.

For (B), If the investor is not diversified (or the portfolio represents the investor's entire wealth), the Sharpe (SPI) is a good measure. Peter's SPI is 0.194 and Betty's is 0.250.

For (C), Betty's portfolio already lies on the efficient frontier because her Sharpe ratio equals the Market's, which is also $6.0\%/24\% = 0.250$; we do not expect portfolios to be located beyond the efficient portfolio.

38. Emily listed the daily returns last week for two cryptocurrencies, bitcoin (BTC) and Ethereum (ETH). These are very small samples of only five returns for each currency. The returns are displayed below along with their ranks; e.g., Wednesday saw the worst performance for both cryptocurrencies, while Tuesday saw the best performance. Albeit a very small sample, the ranks do match on three of the days (Monday, Tuesday, and Wednesday) and differ on only two days (Thursday and Friday).

Day (sort by BTC)	Return (as %) of BTC	Return (as %) of ETH	Rank of BTC	Rank of ETH
Wed	-3.50	-4.70	1	1
Thur	-2.90	0.75	2	3
Fri	1.40	0.30	3	2
Mon	4.70	2.80	4	4
Tue	5.80	9.90	5	5

What are, respectively, the Spearman's rank and Kendall's tau?

A. -0.70 (rank) and -0.55 (Kendall's)

B. 0.33 (rank) and 0.67 (Kendall's)

C. 0.90 (rank) and 0.80 (Kendall's)

D. 1.40 (rank) and 1.60 (Kendall's)

答案:C

解析:

Option C is correct. True: 0.90 (rank) and 0.80 (Kendall's)

The Spearman's rank correlation is given by $1 - \frac{6\times 2}{((5^2-1)\times 5)} = 0.90$. Given there are nine (9) concordant pairs and only one (1) discordant pair, the Kendall's tau is given by $\frac{9-1}{(5\times 4/2)} = 0.80$.

考察: Quantitative Methods: Correlation　重要程度: [■■□]

科目: Quantitative Analysis

39. The spot price of oil is \$100.000 per barrel while the riskfree rate is 3.0% and its storage cost is 2.0% per annum. Oil has positive systematic risk. Specifically, its beta with respect to the market, $\beta(\text{Oil, M}) = 1.10$. The market's expected return is 14.0%; i.e., the market's expected excess return is +11.0%. Finally, the expected growth rate of the oil price (aka, price appreciation) is +8.0%. All rates are per annum with continuous compounding. We are interested in the traded price of the fifteen-month (i.e., 1.25 year) forward contract, and we'll assume the traded price equals the theoretical price. Which of the following statements is true about (i) the slope of the forward curve and (ii) whether we expect normal contango or normal backwardation?

A. The forward curve is inverted, $F(1.25) = \$97.50$, and we perceive normal contango, $F(1.5) > E[S(1.25)]$

B. The forward curve is inverted, $F(1.25) = \$95.00$, and we perceive normal backwardation, $F(1.5) < E[S(1.25)]$

C. The forward curve is upward-sloping, $F(1.25) = \$105.00$, and we perceive normal contango, $F(1.5) > E[S(1.25)]$

D. The forward curve is upward-sloping, $F(1.25) = \$102.50$, and we perceive normal backwardation, $F(1.5) < E[S(1.25)]$

答案:B
解析:
B. True: The forward curve is inverted, $F(1.25) = \$95.00$, and we perceive normal backwardation, $F(1.5) < E[S(1.25)]$
The expected future spot price is given by, $E[S(1.25)] = S(0) * \exp(g * T) = \$100.0 * \exp(8.0\% * 1.25) = \$110.517$; i.e., the current spot projected at its growth rate.
The discount rate is given by $k = 3.00\% + 11.00\% * 1.10 = 15.10\%$; i.e., per the CAMP where $\beta(\text{Oil, M}) = 1.10$. The theoretical futures price is given by $F(1.25) = E[S(1.25)] * \exp[(RF - k) * T] = \$100.517 * \exp[(3.00\% - 15.10\%) * 1.25] = \$95.004$.
考察: Finance: Derivatives **重要程度:** [■■□]
科目: Quantitative Analysis

40. Peter is a Financial Analyst whose boss asked him to develop a model, or if necessary a set of models, that are multi-factor risk models for the firm's fixed income and derivative investment portfolios. Specifically, there are four objectives as follows:
I. To measure and hedge the risk of bond portfolios in terms of the relatively small number of liquid bonds available, in particular, on-the-run government bonds
II. To measure and hedge the risk of a portfolio of swaps in terms of the highly liquid money market and swap instruments, which are more numerous
III. To measure the risk of mixed portfolios not terms of other securities but instead to model direct changes to the shape of the term structure
IV. To measure portfolio volatility by assuming a term structure of volatility but doing so without an excessive "curse of dimensionality;"i.e., avoiding too many correlation pairs
For these four objectives, Peter has three basic multi-factor risk metrics: key-rate exposures, partial '01s, and forward-bucket '01s. For which of the four objectives is key-rate exposures best suited?

A. I. only

B. III. only

C. I. and IV. only

D. I., II., III. and IV.; i.e., all four

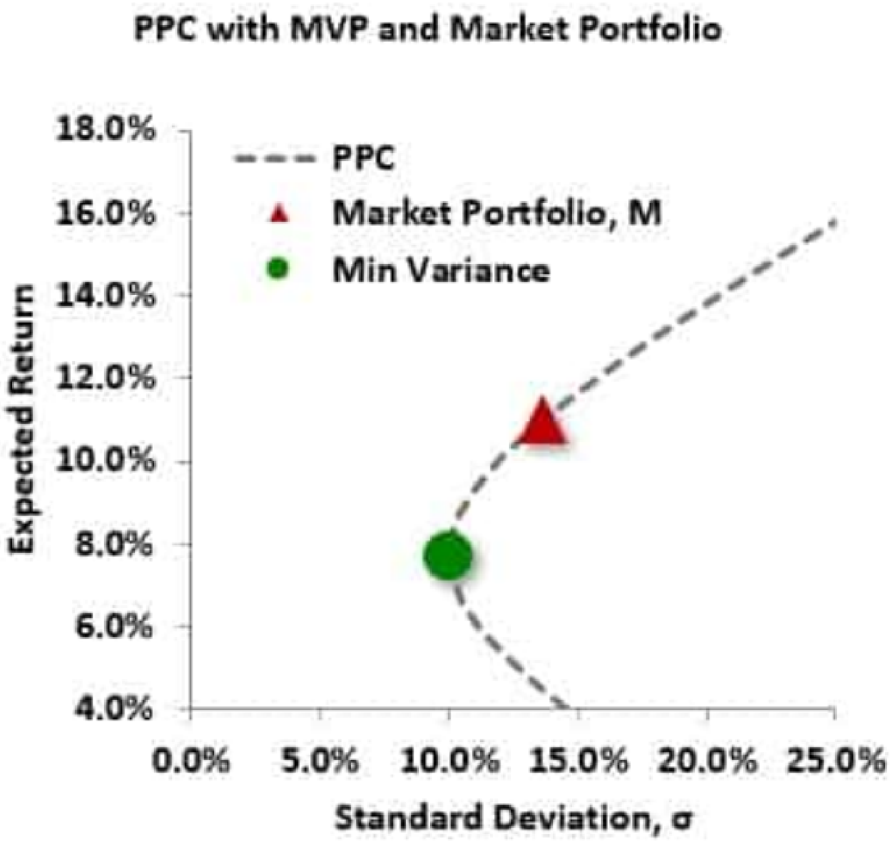
答案:C
解析:
C 选项正确。

- *Key-rate exposures* are used for measuring and hedging the risk of bond portfolios in terms of a relatively small number of the most liquid bonds available, usually the most recently issued, near-par, government bonds.
- *Partial '01s* are used for measuring and hedging the risk of portfolios of swaps or portfolios that contain both bonds and swaps in terms of the most liquid money market and swap instruments. As these instruments are almost always those whose prices are used to build a swap curve, the number of securities used in this methodology is usually greater than the number used in a key-rate framework.

- Finally, *forward-bucket '01s*, mostly used in the swap or combined bond and swap contexts as well, measure the risk of a portfolio not in terms of other securities but in terms of direct changes in the shape of the term structure. As a result, forward-bucket '01s are often the most intuitive way to understand the curve risks of a portfolio, but not the quickest way to see which hedges are required to neutralize such risks. This chapter concludes with a comment on the use of these methods to measure the volatility of a portfolio.

考察: Finance: Fixed Income Analysis　　**重要程度:** [■■□]
科目: Quantitative Analysis

41. The plot below illustrates an actual portfolio possibilities curve (PPC, dashed line). Two prominent portfolios (minimum variance and market portfolio) are also plotted.



The green point is the minimum variance portfolio, MVP, which has an expected return of 7.70% and standard deviation, σ, of 10.0%. The red triangle is the Market Portfolio that has an expected return of 11.0% and standard deviation, σ, of 13.7%. If the riskfree rate is 4.0%, what is the formula for the capital market line (CML)?

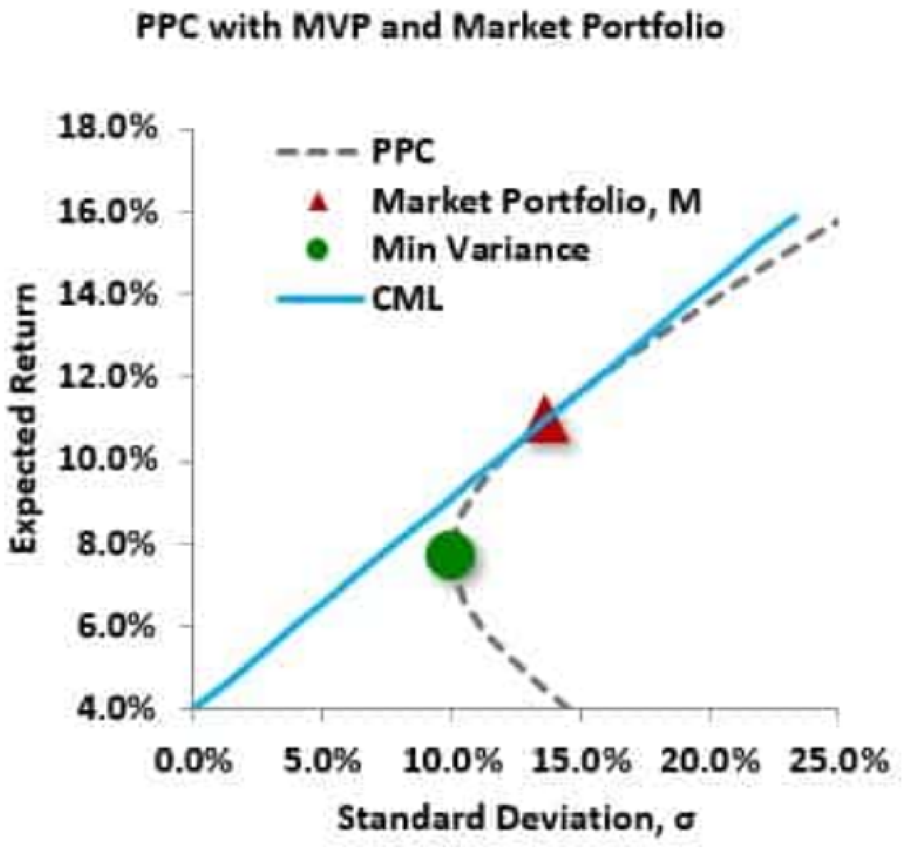
A. $E(R) = 0.040 + 0.511 \times \sigma(P)$

B. $E(R) = 0.137 + 0.511 \times \sigma(P)$

C. $E(R) = 0.040 + 0.375 \times \beta(P, M)$

D. $E(R) = 0.077 + 0.511 \times \beta(P, M)$

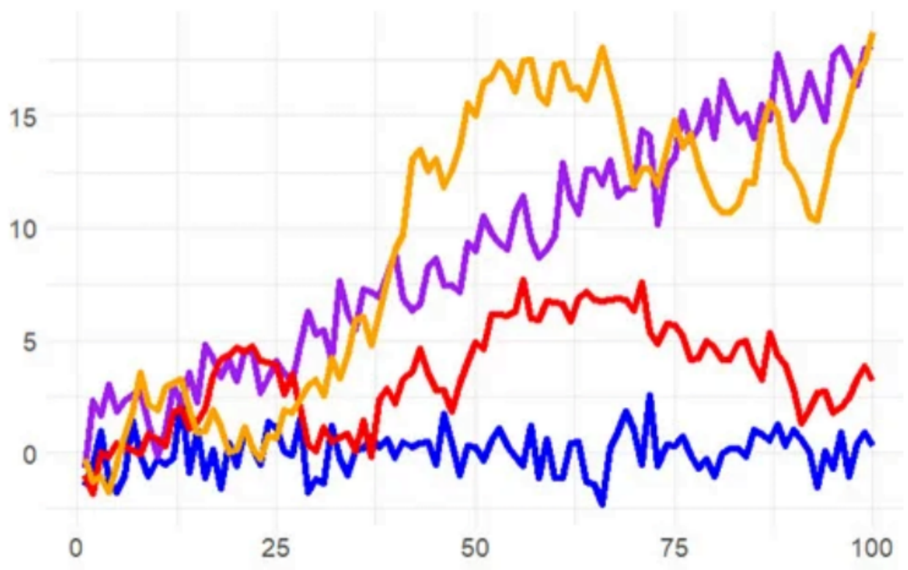
答案:A
解析:
Option A is correct. A. True: the CML is here given by $E(R) = 0.040 + 0.511 \times \sigma(P)$
The capital market line (CML) must have the y-intercept intercept at the riskfree rate, which was given as 4.0%. Its tangent is the market portfolio. The slope of the CML is the market portfolio's Sharpe ratio, which is $(11.0\% - 4.0\%)/13.7\%$. Therefore, the CML is given by $E(R) = 0.040 + (0.110 - 0.040)/0.1370 \times \sigma(P) = 0.040 + 0.511 \times \sigma(P)$.



考察: Finance: Portfolio Management　　**重要程度:** [■■□]

科目: Quantitative Analysis

42. Jeremy simulates four time series: white noise, white noise with trend, a random walk, and a random walk with drift. They are plotted below.



Jeremy conducts an Augmented Dickey-Fuller (ADF) test with the R function `adf.test()`. According to GARP, the ADF is the most widely used unit root test. Its null hypothesis is that the series is a unit root process and its corresponding alternative hypothesis is that the series is stationary. Importantly, this `adf.test()` does allow an intercept and trend: it evaluates a trend-stationary process as stationary but evaluates a difference-stationary process (prior to transformation) as non-stationary. If Jeremy performs an ADF test on each of these four series, what is the most likely outcome?

A. All four have low p-values; e.g., p-values below 0.050

B. All four have high p-values; e.g., p-values above 0.100

C. The random walks have low p-values but the white noise processes have high p-values

D. The white noise processes have low p-values but the random walks have high p-values

答案:D
解析:
D 选项正确。 The white noise processes have low p-values but the random walks have high p-values

43. The current level of a stock index is 7,790 while the risk-free rate is 4.0% per annum with annual compounding. The dividend yield assumption is 1.50% with continuous compounding. We can observe that the nine-month futures contract is currently trading at a price of 8,000. If an index arbitrage is possible, what is the index arbitrage trade?

A. Buy the underlying stocks and short the futures contract

B. Sell the underlying stocks and buy the futures contract

C. Cannot figure because compound frequencies differ

D. The implied reverse cash-and-carry may not be possible if the ease lease rate on the borrowed stocks exceeds the risk-free rate

答案:A
解析:
A 选项正确。 Buy the underlying stocks and short the futures contract. The theoretical price of the futures contract is given by $7,790 \times (1 + 4.0\%)^{0.75} \times \exp(-1.50\% \times 0.75) = 7,932.80$. The observed futures price of 8,000 is trading rich such that the arbitrage is a cash and carry variation: buy the stocks and short the futures contract on the index.
考察: Derivatives: Forwards and Futures　　**重要程度:** [■■□]
科目: Quantitative Analysis

44. On May 28, 2019, Robin the Portfolio Manager considers the purchase of \$100.0 million face amount of the U.S. Treasury 2 3/8s (2.375%) due November 15, 2028, at a cost of $\$90,250,000$. She refers to this as her "bullet" investment. After an analysis of the interest rate environment, she is comfortable with the pricing of the bond at a yield of 3.60% and with its duration of 8.32 years.

Data on Three U.S. Treasury Bonds as of May 28, 2019

Coupon	Maturity	Years	Price	Yield	Duration, D	Convexity
2.500%	3/31/24	4.8	\$102.300	2.00%	4.55	23.6
2.375%	11/15/28	9.5	\$90.250	3.60%	8.32	78.4
4.625%	11/15/48	29.5	\$98.810	4.70%	15.96	365.9

However, upon further inspection, Robin considers an alternative to the above-mentioned bullet investment in the 10-year 2 3/8s. Specifically, the alternative is a "barbell" investment in the shorter maturity 5-year 2 1/2s and the longer maturity 30-year 4 5/8s. The barbell would be constructed to have the same COST and DURATION as the bullet investment. Which of the following is nearest to the convexity of the alternative barbell portfolio?

A. 50.4 years^2

B. 78.4 years^2

C. 136.7 years^2

D. 305.9 years^2

答案:C
解析:
Option C is correct. True: 136.7 years^2

We need to solve for two unknowns simultaneously:

$$V(5) + V(30) = 90.250, \text{ and}$$

$$\frac{V(5)}{90.250} \times 4.55 + \frac{V(30)}{90.250} \times 15.96 = 8.32.$$

Solving, $V(5) = \$60.43$ and $V(30) = \$29.82$, such that the barbell's duration is given by

$$D(\text{barbell}) = \frac{\$60.43}{90.250} \times 4.55 + \frac{\$29.82}{90.250} \times 15.96 = 8.32 \text{ years}$$

(by definition of the solution!); and the barbell's convexity is given by:

$$C(\text{barbell}) = \frac{\$60.43}{90.250} \times 23.6 + \frac{\$29.82}{90.250} \times 365.9 = 136.70 \text{ years}^2$$

考察: Fixed Income　　**重要程度:** [■■□]
科目: Quantitative Analysis

45. Securitization is a trend that is over fifty years old: The Housing and Urban Development Act of 1968 gave birth to Ginnie Mae (https://www.ginniemae.gov/) in an effort to promote homeownership by way of guaranteeing residential mortgage-backed securities (MBS). As GARP explains (see Figure 4.2 of Chapter 4) there have been several milestones along the way. Each of the following statements about securitization is true EXCEPT which is false?

A. Securitization offers at least some diversification of credit risk

B. Securitization transfers a substantial amount of risk, including but not limited to credit risk, from the originator's balance sheet to investors

C. The 2007-2009 global financial crisis (GFC) discredited the principles of securitization as evidenced by the subsequent evaporation of asset-backed securities (ABS)

D. Securitization enables originate-to-distribute (OTD) but the 2007-2009 crisis witnessed poor securitization risk management that allowed too much systemic concentration of risk

答案:C
解析:
C 选项正确。 The crisis neither discredited securitization nor saw the evaporation of asset-backed securities.
In regard to (A), (B) and (D), each is TRUE.
The first two true statements (i.e., A and B) capture the two essential features of classical securitization:

- by pooling credit-sensitive assets, it provides diversification; and
- By issuing liabilities (notes) to investors, it transfers credit risk.

考察: Credit Risk: Securitization　　**重要程度:** [■■□]
科目: Quantitative Analysis

46. Consider the following seasonal time series model that is represented with lag notation:

$$\boldsymbol{(1 - \phi_1 L)(1 - \phi_S L^{12})Y_t = \delta + \varepsilon_t}$$

$$\downarrow \qquad \qquad \downarrow$$

$$\boldsymbol{(1 - 0.53L)(1 - 0.26L^{12})Y_t = 0.17 + \varepsilon_t}$$

Note that $\phi(1) = 0.530$ and $\phi(s) = 0.260$. In regard to this seasonal model, each of the following statements is true EXCEPT which is false?

A. This seasonal model is a restricted AR(13) model

B. The coefficient on lag 13 is -0.13780

C. The coefficients on lags 2 through 11 are zero

D. This seasonal model is deterministic because lag 12 is a dummy variable

答案:D
解析:
D 选项错误。虚拟变量是分类的 0 或 1，但这是一个随机（且平稳）的 ARMA 模型，其系数 0.26 对滞后 12 项加权；因此，这是 ARMA 模型的一个特例，可以用符号表示为 $\text{ARMA}(1,0) \times (1,0)_{12}$。
关于 (A)、(B) 和 (C)，每个都是正确的。具体而言：这个季节性模型是一个受限的 AR(13) 模型。滞后 13 的系数为 -0.13780，因为 $[1-\phi(1)L] \times [1-\phi(s)L^{12}] \times Y(t) = \delta + \varepsilon(t) \rightarrow [1-\phi(1)L-\phi(s)L^{12}+\phi(1)\times\phi(s)L^{13}] \times Y(t) = \delta+\varepsilon(t) \rightarrow Y(t) = \delta+\phi(1)\times Y(t-1)+\phi(s)\times Y(t-12)-\phi(1)\times\phi(s)\times Y(t-13)+\varepsilon(t)$。滞后 2 到 11 的系数为零。
考察: Quantitative Methods: Time Series　　**重要程度:** [■■□]
科目: Quantitative Analysis

47. The current spot exchange rate between the euro (EUR) and quote currency YYY is 1.3000 EURYYY. The EUR risk-free rate is 4.0% and the YYY risk-free rate is 2.0%; each interest rate is per annum with annual compounding. According to covered interest rate parity (CIRP), which is nearest to the forward rate in nine months ($T = 0.75$ years) quoted in points?

A. -12,576

B. -188.0

C. +33.0

D. +291.0

答案:B
解析:
B 选项正确。 -188.0 forward rate points
According to CIRP, 1.30 EURYYY $\times$ $[(1.02/1.04)]^{(9/12)} = 1.28120$, and $(1.28120 - 1.3000) \times 10,000 = -188.0$.
考察: Finance: Foreign Exchange　　**重要程度:** [■■□]
科目: Quantitative Analysis

48. A speculative (aka, junk) bond has 15.0 years to maturity and pays a semi-annual 6 1/8 coupon; i.e., its coupon rate is 6.125% payable semi-annually. Its yield is 11.00%. If we assume a reasonable shock value (i.e., less than 100 basis points), which of the following is **nearest** to the bond's effective convexity?

A. 12.4 years^2

B. 98.0 years^2

C. 124.3 years^2

D. Cannot answer because face value is not given

答案:B
解析:
B 选项正确。98.0 years^2 $= (1/\$64.574) * [(\$61.996 + \$67.310 - 2 * \$64.574)/0.0050^2]$ where we've selected 50 basis points as the shock. In this case, the initial price $= -\text{PV}(0.1100/2, 15*2, 100*0.06125/2, 100) = \$64.574$, and if the yield shock is $+/-$ 50 bps, then:

- $\text{Price}(\text{yield} + 0.0050) = -\text{PV}(0.1150/2, 15*2, 100*0.06125/2, 100) = \$61.996$, and
- $\text{Price}(\text{yield} - 0.0050) = -\text{PV}(0.1050/2, 15*2, 100*0.06125/2, 100) = \$67.310$.

In regard to false (D), please note that the convexity will be unchanged if we (consistently) change the face value to any amount. Further, while the convexity will vary very slightly as we alter the shock value, it is approximately 98.0 years for shock values below 1.0%.

考察: Fixed Income: Bond Convexity　　**重要程度:** [■■□]

科目: Quantitative Analysis

49. In regard to limits policies, GARP explains that "optimal risk governance requires the ability to link risk appetite and limits to specific business practices. Accordingly, appropriate limits need to be developed for each business as well as for the specific risks associated with the business (as well as for the entire portfolio of the enterprise)." Most institutions set two types of limits, tier 1 (one) and tier 2 (two) limits. About these limits, which of the following is TRUE?

A. Firms should choose and either adopt tier 1 or tier 2 limits but not both simultaneously

B. Tier 1 (tier one) limit exceedances must be cleared or corrected immediately, while tier 2 (tier two) exceedances are less urgent and can be cleared within a few days or a week.

C. Tier 2 (tier two) limits are specific and often include an overall limit by asset class, an overall stress-test limit, and a maximum drawdown limit

D. Tier 1 (tier one) limits are more generalized and relate to areas of business activity as well as aggregated exposures categorized by credit rating, industry, maturity, and region

答案:B

解析:

B 选项正确。 Tier 1 (tier one) limit exceedances must be cleared or corrected immediately, while tier 2 (tier two) exceedances are less urgent and can be cleared within a few days or a week.

Explains GARP, "Most institutions set two types of limits. 1. Tier 1 limits are specific and often include an overall limit by asset class, an overall stress-test limit, and a maximum drawdown limit. 2. Tier 2 limits are more generalized and relate to areas of business activity as well as aggregated exposures categorized by credit rating, industry, maturity, and region, and so on."[1]

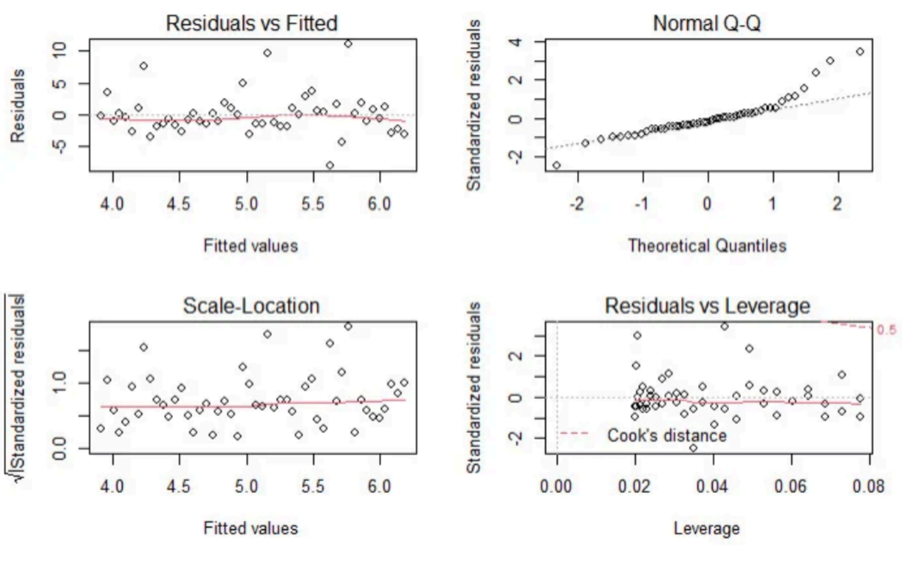
In regard to (A), (C) and (D), each is FALSE.

[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions

考察: Foundations of Risk Management　　**重要程度:** [■■□]

科目: Quantitative Analysis

50. Patrick generated a simple regression line for a sample of 50 pairwise observations. After generating the regression model, he ran R's built-in plot(model) function which produces a standard set of regression diagnostics. These four plots are displayed below.



About these diagnostic plots, which of the following statements is TRUE?

A. There are many outliers

B. The data is significantly heteroskedastic

C. The residuals are a bit heavy-tailed (non-normal) on the right side

D. The residuals reveal that the relationship between the explanatory and response variable is non-linear

答案:C

解析:

Option C is correct. The residuals are a bit heavy-tailed (non-normal) on the right side.

In regard to (A), (B), and (D) each is false. We do see potential outliers in the Residual vs Fitted plot, however the Residuals vs. Leverage plot shows there are no observations with a Cook's distance greater than 0.5 (i.e., no observations above the dotted red line). Heteroskedasticity is not demonstrated, and the plots do support an approximately linear relations. In regard to the specific plots:

- *Residuals vs Fitted*: This plots residuals against the fitted values. We would like to see the residuals randomly scattered across the zero (which these are). The scatter pattern is relatively even suggesting homoskedasticity; i.e., we do not see a pattern that suggests heteroskedasticity. There are not many outliers. This is pretty good-looking residuals vs fitted plot suggestive of a decent linear regression.

- *Normal Q-Q*: If the distribution is normal, the plot will approximate along the straight line. But notice how this plot contains an obvious heavy-tail on the right side.

- *Scale-location*: This plot is similar to the Residuals vs Fitted plot, but the residuals are standardized. It is also used to evaluate heteroskedasticity. But, again, we do not perceive strong evidence of a non-constant variance.

- *Residuals vs Leverage*: The red dashed line represents a Cook's distance of 0.5, but there are not observations outside of this line (i.e., in the upper-left) such that we do not have a case for outlier(s).

考察: Machine Learning: Linear Regression　　**重要程度:** [■■□]

科目: Quantitative Analysis

51. On January 15th of Year 1, a company decides to hedge the planned purchase of 100,000 bushels of corn thirteen months later (on February 15 of Year 2). Below are displayed the per-bushel futures prices of three selected contracts on four different dates.

Bushels	**100,000**
Per Contract (Size)	**5,000**

	Year 1			Year 2
	Jan-15	Apr-15	Aug-15	Feb-15
May, Year 1 Futures Price	$5.85	$6.00		
Sep, Year 1 Futures Price		$6.10	$6.30	
March, Year 2 Futures Price			$6.40	$6.50

Assume the following: at the time of ultimate purchase (February 15 of Year 2), the spot price is $6.47 per bushel when the final contract's basis is $\$6.47 - \$6.50 = -\$0.03$. If the company employs a stack-and-roll strategy, what is the company's net (i.e., after hedging) cost to acquire the corn? [inspired by GARP's EOC Question 8.20]

A. $\$45,000$

B. $\$602,000$

C. $\$647,000$

D. $650,000

答案:B

解析:

B 选项正确。 In this scenario, because the company plans to purchase corn in the future, the hedge is a stack-and-roll of long futures contracts:

- The May of Year 1 contracts are bought at $5.85 and sold at $6.00 in April of Year 1.
- The September of Year 1 contracts are bought at $6.10 sold at $6.30 in August of Year 1.
- The March of Year 2 contracts are bought at $6.40 and sold at $6.50 in February of Year 2.

The gain per contract is $5,000 * [(\$6.00 - \$5.85) + (\$6.30 - \$6.10) + (\$6.50 - \$6.40)] = 5,000 * \$0.45 = \$2,250.00$; and given 20 contracts are used (i.e., 100,000 bushels divided by 5,000 per contract), the total gain on the stack-and-roll hedge is $20 * \$2,250.00 = \$45,000$; put another way, $0.45 per bushel * 100,000 bushels $= \$45,000.00$. The net cost is therefore $(\$6.47 * 100,000) - \$45,000.00 = \$602,000.00$

考察: Financial Markets and Products　　**重要程度:** [■■□]

科目: Financial Markets and Products

52. Consider a 9-year bond with a semi-annual 10.0% coupon that has a current price of $119.780 and a yield of 7.000%. If the yield drops to 6.850%, the bond's price increases to $120.900. If this is the case, then which of the following is **nearest** to the bond's dollar value of '01 (DV01)?

A. -0.0121

B. 0.0121

C. 0.0747

D. 1.1200

答案:C

解析:

C 选项正确。 True: DV01 = 0.0747. The price change is $\$120.900 - \$119.780 = \$1.120$ given a 15-basis point drop in the yield, which is $\$1.120/15 = 0.0746667$ per basis point. Note that strictly (technically) the DV01 is the price change associated with a decrease of one basis point. This will be an almost identical price change associated with an increase of one basis point, but it won't be exactly the same.

考察: Fixed Income: Risk Measures　　**重要程度:** [■■□]

科目: Quantitative Analysis

53. GARP explains that "risk management must be implemented across the entire enterprise under a set of unified policies and methodologies. (This is called enterprise risk management ...). The infrastructure of risk management, which includes both physical resources and clearly defined operational processes, must be up to the task of an enterprise-wide scope."[1] However, it is a difficult and daunting task to evaluate the fitness of a firm or bank's risk management system. Among the following features, indicators, or characteristics, which is MOST LIKELY to signify that the firm is serious about its risk management process?

[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions, 10/2019.

A. The firm's risk manager(s) is a well-compensated member of the executive staff with compelling career opportunities

B. At least one-third of executive compensation plans consist of stock options, or in the case of private firms, phantom stock options

C. The board has quantified the firm's agency costs and can show that agency costs are within a quantifiable confidence interval, or at least below an upper numeric bound

D. The board's audit committee reports to the board's risk management committee and every member of the board is also a member of the board's risk management committee

答案:A

解析:

A 选项正确。 The firm's risk manager(s) is a well-compensated member of the executive staff with compelling career opportunities.

We think GARP makes an excellent point that a true test of the firm's view of its risk managers is revealed by how it treats the human capital. Explains GARP, "One way to measure the seriousness of a risk management process is to examine the human capital employed and the risk managers'standing within the corporate hierarchy: Is the risk manager considered to be a member of the executive staff and can this position lead to other career opportunities? How independent is the risk manager? What authority does he or she hold? To whom does he or she report? Are risk managers paid well relative to other employees who are rewarded for performance (e.g., traders)? To what extent can one characterize the enterprise's ethical culture as being strong and resilient against the actions of bad actors? Has the firm set clear-cut ethical standards and are these standards actively enforced?"[1]

In regard to (B), (C) and (D), these are false!

[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions, 10/2019.

考察: Foundations of Risk Management　　**重要程度:** [■■□]

科目: Quantitative Analysis

54. Mary works for an insurance company and she has regressed medical costs (aka, the response or dependent variable) for a sample of patients against four independent variables: AGE, BMI, SMOKER, and CHARITY. The sample's average age is 38.51 years. Body mass index (BMI) is mass divided by height squared and the sample's average BMI is 22.16. SMOKER is a dummy variable where zero indicates a non-smoker and 1 indicates a smoker; the sample's average SMOKER value is 0.163 which indicates that 16.3% of the sample are smokers. CHARITY is the dollar amount of charitable spending in the last year; the sample average is $490.70 donated to charity in the last year. Mary's regression results are displayed below.

Medical COST regressed against AGE + BMI + SMOKER(1/0) + CHARITY($)
Simulated dataset

Coefficient	Estimate	Std Error	t-stat	p value
(Intercept)	**−324.21**	**415.77**	**−0.78**	**4.40×10^{-1}**
AGE	**56.95**	**7.55**	**7.54**	**4.60×10^{-9}**
BMI	**111.76**	**11.94**	**9.36**	**2.06×10^{-11}**
SMOKER	**454.08**	**125.83**	**3.61**	**8.84×10^{-4}**
CHARITY	**−0.55**	**0.18**	**−3.06**	**4.06×10^{-3}**

Residual standard error: 295.2 on 38 degrees of freedom
Multiple R-squared: 0.8343, Adjusted R-squared: 0.8168
F-statistic: 47.82 on 4 and 38 DF, p-value: 2.486e-14

Each of the following statements is true about these regression results EXCEPT which is false?

A. The sample size is 43 patients

B. Mary can reject a null hypothesis that all explanatory variables (jointly) have zero coefficients

C. Mary can infer that patient medical cost is positively associated with each of AGE, BMI, and, on average, is greater for a smoker

D. Mary should suspect problematic multicollinearity because the intercept is suspiciously negative and the adjusted R-squared is too near to the unadjusted R-squared

答案:D
解析:
D 选项正确。False (triple false!). The telltale symptom of problematic multicollinearity is a high R^2 with few significant t-stats (but all t-stats are significant). The negative intercept is not itself suspicious: it is simply out-of-sample. Finally, given $n = 43$ and $k = 4$, per adjusted $R^2 = 1 - [(1 - R^2) * (n - 1)/(n - k - 1)]$, the adjusted R^2 is by definition near to the unadjusted R^2. In regard to (A), (B) and (C), each is TRUE

- The sample size is 43 patients
- Mary can reject a null hypothesis that all explanatory variables (jointly) have zero coefficients
- Mary can infer that patient medical cost is positively associated with each of AGE, BMI and, on average, is greater for a smoker

考察: Quantitative Methods　　**重要程度:** [■□□□□]
科目: Quantitative Analysis

55. Today an exchange introduced a new futures contract. The open interest started at zero. Because it is a new contract, there were only the following four trades:

- Trader Adam (TA) buys (aka, takes a long position in) 1,000 contracts from Trader Betty (TB) who is the seller (aka, takes a short position) and both are new positions
- Trader Charles (TC) buys 300 contracts, in a new position, from Trader Adam (TA) who closes out 300 of his contracts
- Trader Denise (TD) sells 225 contracts, in a new position, to Trader Betty (TB), the buyer who closes out 225 of her contracts
- Trader Betty (TB) delivers on 175 of her short contracts to Trader Adam (TA) who takes delivery on 175 of his long contracts

What is the open interest, after the above four trades, at the end of the day?

A. Zero

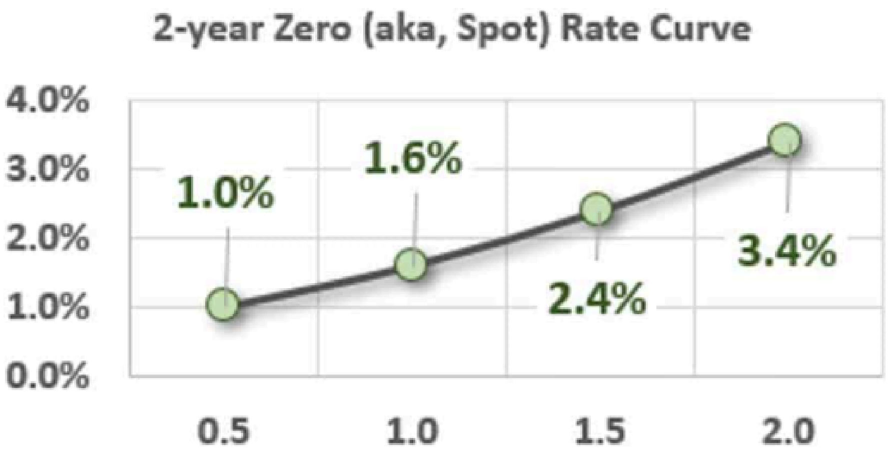
B. 300 contracts

C. 825 contracts

D. 1,000 contracts

答案:C
解析:
C 选项正确。 TRUE: 825 contracts
考察: Futures Markets　　**重要程度:** [■■□]
科目: Quantitative Analysis

56. Assume that the following upward-sloping zero rate (aka, spot rate) curve prevails:



Tuckman introduces the concept of a coupon effect in Chapter 3. Each of the following statements relates to this coupon effect and is necessarily true EXCEPT which statement is false?

A. Given this spot rate curve, the two-year par rate (aka, par yield) must be less than 3.4%

B. Given this spot rate curve, a zero-coupon bond with a two-year maturity must have a yield (yield to maturity) of 3.4%

C. Given this spot rate curve, a coupon-bearing bond with a two-year maturity must have a yield (yield to maturity) that is less than 3.4%

D. Given this spot rate curve, the yields (yields to maturity) of all bonds with two-year maturities must be identical due to the Law of One Price

答案:D
解析:
Option D is correct. The Law of One Price effectively requires that there is only one discount factor at each maturity, which is equivalent to saying that each maturity can have only one zero rate.
This statement (D) is false precisely because of the coupon effect: the yields of two-year bonds will vary as their coupons vary. Yield is a complex average of all the relevant spot rates and, importantly, is a function of the bond's cash flows.
In regard to (A), (B), and (C), each is TRUE. In regard to true (C), the 2-year par yield implied by this zero rate curve is 3.3621% because a 2-year bond with a coupon rate of 3.3621% has a theoretical price of exactly $100.00 under this curve, and therefore it's yield is also 3.3621%.
考察: Finance: Fixed Income Analysis　　**重要程度:** [■■□]
科目: Quantitative Analysis

57. To hedge risk, the firm's toolbox includes derivatives such as swaps, futures, forwards, and options. About the use of derivatives to hedge, each of the following is true EXCEPT which is inaccurate?

A. Derivatives represent a decision to transfer (or mitigate) the risk(s) when the firm neither wants to retain nor avoid the risk

B. Because hedging can only mute the volatility of accrual-based earnings, hedging cannot increase (or even alter) the firm's cash flows and therefore cannot influence agency risks

C. A large conglomerate (i.e., a firm with multiple business units operating in different industries/sectors) is more likely to create natural hedges than a small, focused firm

D. Airlines can cross-hedge the cost of jet fuel with futures contracts on (crude or heating) oil, but if the commodity's price drops rapidly, unhedged airlines are likely to outperform their hedged competitors

答案:B
解析:
B 选项正确。 Inaccurate. Hedging can alter cash flows and can leverage agency risks

Explains GARP, "[H]edging can make sense for investors if it is used as a tool to increase the firm's cash flows (rather than to reduce equity investor risk). For example, firms may need to offer their customers a stable price over the next three years, which may be impossible without hedging a key cost input. If hedging like this increases customer demand, then equity investors are happy ... Equity investors are also happy if the firm uses hedging to reduce its tax bill (e.g., by stabilizing revenues from one year to the next). Again, hedging has the effect of increasing after-tax revenues. Finally, equity investors are not the only stakeholders, and certainly not the only decision-makers. Managers, regulators, and general staff expect the firm to be financially sound and protected from sudden mishaps. Less legitimately, managers may use hedging to ensure their firm meets key short-term targets (e.g., stock analyst expectations) that affect their prestige and compensation. Risk managers need to pay close attention to how derivatives can leverage agency risks."[1]
In regard to (A), (C), and (D), each is TRUE
[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions, 10/2019
考察: Foundations of Risk Management **重要程度:** [■■□]
科目: Quantitative Analysis

58. Debra is an economist who is interested in the relationship between consumer spending and the gross domestic product (GDP). From the FRED database at the Fed's Bank of St. Louis (https://fred.stlouisfed.org/) she collects quarterly data from 1980 through the first quarter of 2020; her series includes $n = 161$ quarters of data. She regresses consumer spending (C_SPEND), as the response (aka, dependent) variable against GDP as the explanatory (aka, independent) variable. Each series is not a level, but rather a seasonally adjusted percent change. The regression results are displayed below.

Consumer Spending (C_SPEND) regressed against Gross Domestic Product (GDP)
Quarterly Growth (Seasonally adjusted), 1980 to 2020:Q1, n = 161

Coefficient	Estimate	Std Error	t-stat	p value
(Intercept)	**0.9000**	**0.3156**	**2.8520**	**4.92×10^{-3}**
GDP	**0.3659**	**0.0499**	**7.3285**	**1.11×10^{-11}**

Source: FRED at https://fred.stlouisfed.org/

Additionally, Debra calculates the (univariate) standard deviation of each variable over the period: σ(C_SPEND) = 2.484 and σ(GDP) = 3.412. In regard to these regression results, each of the following statements is true EXCEPT which is false?

A. The coefficients are jointly insignificant because each two-sided 95.0% confidence interval contain zero

B. The GDP's t-statistic ("t-stat") of 7.3285 is equal to the coefficient estimate (0.3659) divided by its standard error (0.0499)

C. Given the variables' cross-volatility, $\sigma(\text{C\_SPEND})/\sigma(\text{GDP}) = 2.484 \div 3.412 = 0.7280$, the correlation between the variables is approximately 0.50

D. The coefficient of determination (R-squared; aka, R^2) is about 0.25 such that we can say about 25% in the variation in consumer spending (C_SPEND) is explained by gross domestic product (GPD)

答案:A
解析:
A 选项正确。 False: Each (aka, both) coefficients are SIGNIFICANT (as we can immediately observe by their small p-values) such that neither two-sided 95.0% confidence interval contains zero.
At 95.0% confidence, the CI for the intercept coefficient is given by $0.9000 \pm 1.96 \times 0.3156 = (0.281, 1.518)$. At 95.0% confidence, the CI for the beta (aka, slope coefficient) is given by $0.3659 \pm 1.96 \times 0.0499 = (0.268, 0.464)$.
In regard to (B), (C) and (D), each is TRUE. Specifically:

- True: The GDP's t-statistic ("t-stat") of 7.3285 is equal to the coefficient estimate (0.3659) divided by its standard error (0.0499)

- True: Given the variables' cross-volatility, $\sigma(\text{C\_SPEND})/\sigma(\text{GDP}) = 2.484 \div 3.412 = 0.7280$, the correlation between the variables is approximately 0.50. Because $\beta(\text{C\_SPEND, GDP}) = \rho(\text{C\_SPEND, GPD}) \times \sigma(\text{C\_SPEND})/\sigma(\text{GDP})$, it follows that $\rho(\text{C\_SPEND, GPD}) = \beta(\text{C\_SPEND, GDP}) \times \sigma(\text{GDP})/\sigma(\text{C\_SPEND})$; in this case, $\rho(\text{C\_SPEND, GPD}) = 0.3659 \times 3.412/2.484 = 0.5026$ or about 0.50.

- True: The coefficient of determination (R^2) is about 0.25 such that we can say about 25% in the variation in consumer spending (C_SPEND) is explained by gross domestic product (GPD). In a univariate regression, the R^2 is the square of the correlation coefficient; in this case, the $R^2 = 0.50^2 = 0.25$ (indeed the dataset's regression produces an exact R-squared of 0.2525, and an adjusted R-squared of 0.2478, both of which round to 0.25).

考察: Quantitative Methods: Linear Regression **重要程度:** [■■□]
科目: Quantitative Analysis

59. Derivatives can be cleared through a central counterparty (CCP) or they can be cleared bilaterally; i.e., uncleared transactions are cleared bilaterally. If the derivatives are cleared through a CCP, they can be either exchange-traded or over-the-market (OTC). In this way, we distinguish between the exchange and the clearinghouse (which may be owned by an exchange). While exchanges operate CCPs, we can also refer to the emergent OTC CCPs. In regard to central counterparties (CCPs), which of the following statements is TRUE?

A. Banks are easier to regulate than both OTC CCP and exchange CCP

B. An exchange CCP is more dependent on valuation models than an OTC CCP

C. CCP member defaults are negatively correlated such that CCPs are counter-cyclical

D. In the cleared market, CCPs want initial margin that covers a 5-day 99.0% price value at risk (VaR)

答案:D
解析:
D 选项正确。 In the cleared market, CCPs want initial margin that covers a 5-day 99.0% price value at risk (VaR)
In regard to (A), (B), and (C), each is FALSE. Instead, the following are true statements:

- CCPs are easier to regulate than banks
- An OTC CCP is more dependent than an exchange CCP on valuation models
- CCP member defaults are positively correlated and CCPs are pro-cyclical

考察: Clearing and Settlement **重要程度:** [■■□]
科目: Quantitative Analysis

60. Exactly one year ago, Sally purchased a \$100.00 face value bond that pays a semi-annual coupon with a coupon rate of 9.0% per annum. When she purchased the bond, it had a maturity of 10.0 years and its yield to maturity (YTM; aka, yield) was 6.00%. If the bond's price today happens to be unchanged from one year ago (when she purchased the bond), which of the following is nearest to the bond's yield (yield to maturity) today?

A. 5.57%

B. 5.78%

C. 6.00%

D. 6.22%

答案:B
解析:
B 选项正确。5.78% is the yield to maturity if the bond's price remains unchanged, but its maturity shortens to 9.0 years. When Sally initially purchased the bond its price was equal to $= -PV(0.060/2, 10*2, \$100*0.090/2, 100) = \$122.3162$. If the price is

unchanged, then one year later (i.e., $N-2$ periods), the yield is given by $RATE(9*2, \$100*0.090/2, -122.3162, 100)*2 = 5.7843\%$. Using the TI BA II+ calculator: 20 N, 3 I/Y, 4.5 PMT, 100 FV and CPT $PV = -122.3162$, then we only need to change the periods with 18 N, CPT I/Y [display: $I/Y = 2.8922$] $\times 2 = 5.7843\%$.

考察: Fixed Income   **重要程度:** [■■□]

科目: Quantitative Analysis

61. One of the risk management building blocks is the need to balance risk and reward. Specifically, GARP says, "Economic capital provides the firm with a conceptually satisfying way to balance risk and reward. For each activity, firms can compare the revenue and profit they are making from an activity to the amount of economic capital required to support that activity." Each of the following statements is true about RAROC EXCEPT which is inaccurate?

A. For an activity to increase shareholder value, its RAROC should be higher than the cost of equity capital

B. Four applications of RAROC include business comparison, investment analysis, pricing strategies, and risk management cost/benefit analysis

C. Advantages of RAROC include (i) it has one universal regulatory definition (without credible variants), such that benchmarking against peers is easy; and (ii) it is easy to implement in practice

D. If RAROC's denominator is economic capital, which is typical, then its numerator should be an after-tax risk-adjusted expected return where the risk-adjusted refers to an adjustment for expected losses

答案:C

解析:

C 选项正确。 C. FALSE, doubly. Instead, RAROC is notorious for the lack of a universal definition and several variations (says GARP: "There are many variants on the RAROC formula, applied across many different industries and institutions."). Also, RAROC is not easy to apply: "There are many practical difficulties in applying RAROC, including its dependence on the underlying risk calculations. Business lines often dispute the validity of RAROC numbers, sometimes for self-interested reasons,"explains GARP.

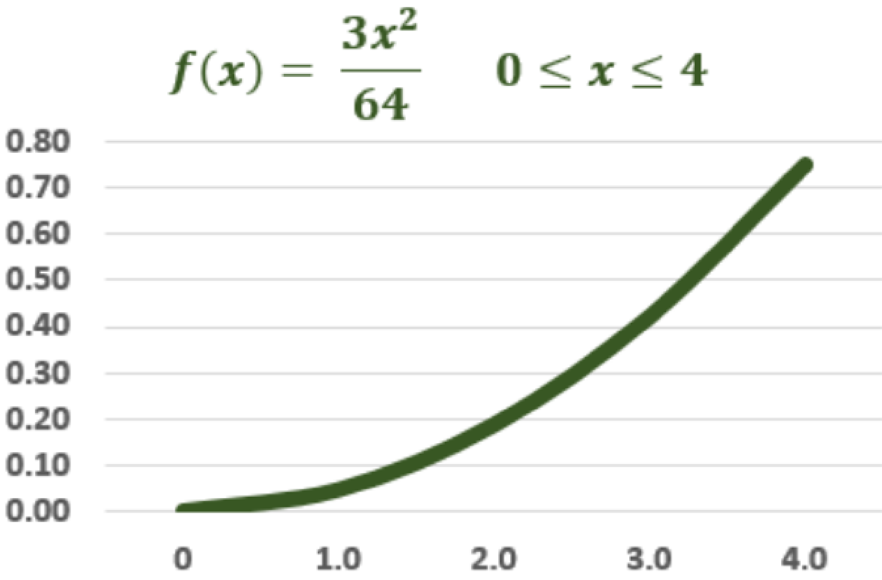
In regard to (A), (B) and (D), each is TRUE. Specifically, each of the following is TRUE:

- For an activity to increase shareholder value, its RAROC should be higher than the cost of equity capital
- Four applications of RAROC include business comparison, investment analysis, pricing strategies, and risk management cost/benefit analysis
- If RAROC's denominator is economic capital, which is typical, then its numerator should be an after-tax risk-adjusted expected return where the risk-adjusted refers to an adjustment for expected losses

考察: Foundations of Risk Management: RAROC   **重要程度:** [■■□]

科目: Quantitative Analysis

62. Consider a random variable that represents the loss severity of a risky asset and is given by the continuous probability distribution $f(x) = 3x^2/64$ on the domain from zero to four:



What are this variable's mean and variance?

A. $\mu = 1.0$ and $\sigma^2 = 3.50$

B. $\mu = 2.0$ and $\sigma^2 = 2.75$

C. $\mu = 3.0$ and $\sigma^2 = 0.60$

D. $\mu = 3.6$ and $\sigma^2 = 0.25$

答案:C
解析:
C 选项正确。 Correct: $\mu = 3.0$ and $\sigma^2 = 0.60$

- To retrieve the mean μ, we integrate $x \cdot f(x)$
- To retrieve the variance, we integrate $(x-\mu)^2 \cdot f(x)$

考察: Quantitative Methods: Probability Distributions **重要程度:** [■■□]
科目: Quantitative Analysis

63. A trader shorts 500 shares when the stock price is \$70.00. The initial and maintenance margin, respectively, are 150.0% and 125.0%. At what share price will further margin be required?

A. \$65.00

B. \$72.00

C. \$79.00

D. \$84.00

答案:D
解析:
Option D is correct. Because the initial margin is 150.0%, the trader is required to contribute $150.0\% \times 500 \times \$70.00 = \$52,500.00$ which is comprised of $\$35,000.00$ from selling the shares plus $\$17,500.00$. If the share price increases to X, the maintenance margin becomes $500 \times 1.25 \times X = 625 \times X$. This is greater than the $\$52,500.00$ when $625 \times X > \$52,500.00$, or when $X > \$52,500.00/625 = \$84.00$.
考察: Finance: Margin Trading **重要程度:** [■■□]
科目: Quantitative Analysis

64. As a market maker, Toughgreen Financial has written (aka, taken a short position in) a single contract for 100 call options on a non-dividend-paying stock whose volatility is 36.0% per annum when the riskless rate is 3.0%. Their strike price is \$100.00 but the options are underwater because the current stock price is \$90.00. The option term is six months. At the current stock price, each option has a value of \$5.88 and each option's percentage delta, $\Delta = +0.410$. From Toughgreen's perspective, if the stock price increases by +\$3.00 to \$93.00, which is nearest to the impact on the position's value?

A. a) A loss of few dollars less than \$8.00; i.e., $100 * [\$5.88 * (\$3.00/\$90.00) * 0.410]$ minus (-) gamma adjustment

B. b) A loss of few dollars more than \$8.00; i.e., $100 * [\$5.88 * (\$3.00/\$90.00) * 0.410]$ plus (+) gamma adjustment

C. c) A loss of few dollars less than \$123.00; i.e., $100 * (\$3.00 * 0.410)$ minus (-) gamma adjustment

D. d) A loss of few dollars more than \$123.00; i.e., $100 * (\$3.00 * 0.410)$ plus (+) gamma adjustment

答案:D
解析:
D. True: A loss of few dollars more than \$123.00; i.e., $100 * (\$3.00 * 0.410)$ plus (+) gamma adjustment.
As a partial first derivative, delta is the rate of change of the option price with respect to the price of the underlying asset (stock, in this case): $\Delta = \partial c/\partial S$. A percentage delta of 0.410 signifies the expectation of a linear change of \$0.410 in the call option value for each \$1.00 change in the stock price. However, this excludes the gamma adjustment: the actual price increase in the call option is GREATER than the change estimated by the linear approximation. Consequently, the short position (who writes the call) is exposed to a greater price drop.
Specifically, in this example (although the problem does NOT require calculating gamma!), the percentage gamma is about 0.0170 such that the loss due to a +\$3.00 stock price change is $\$3.00 * 0.410 + 0.5 * 0.0170 * \$3.00^2 = \$1.3065$. Indeed, the actual option price increases from \$5.882 to \$7.188, for an exact difference of \$1.3060. In this way, the actual loss on the position (of 100 options) is \$130.60 while the delta-gamma estimate is \$130.65
考察: Finance: Option Greeks　**重要程度:** [■■□]
科目: Quantitative Analysis

65. According to GARP, one of the building blocks in risk management is a proper understanding of the difference between expected loss, unexpected loss, and extreme risk; aka, tail risk. In regard to this building block, which of the following statements is TRUE?

A. Effective risk management should reduce a credit portfolio's expected loss (EL) to approximately zero

B. Expected loss is a product of (i) the probability of the risk event occurring; (ii) the severity of the loss if the risk event occurs, and (iii) the expected recovery rate

C. While expected loss (EL) is a function of default correlation, unexpected loss (UL) is NOT influenced by portfolio granularity

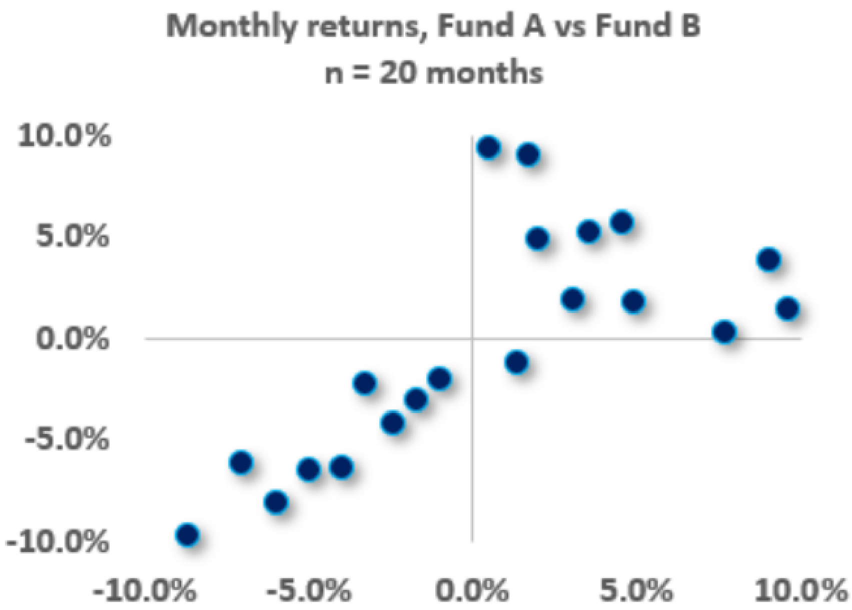
D. Although banks theoretically do not need to set aside provisions when loan products are accurately priced, in realistic practice, banks should provision for expected losses

答案:D
解析:
D 选项正确。 TRUE: Although banks theoretically do not need to set aside provisions when loan products are accurately priced, in realistic practice, banks should provision for expected losses.
Each of (A), (B), and (C) is FALSE. Instead, the following are true:

- All credit portfolios have non-negative expected loss (EL)
- Expected loss is a product of (i) the probability of the risk event occurring; (ii) the severity of the loss if the risk event occurs, and (iii) exposure at default.
- While unexpected loss (UL) is a function of default correlation, expected loss (EL) is not influenced by portfolio granularity

考察: Foundations of Risk Management　　重要程度: [■■□]
科目: Quantitative Analysis

66. Below is a scatterplot of monthly returns, Fund A versus Fund B, over twenty months ($n = 20$).



Skewness is a trivial (special) case of coskewness; e.g., $S(AAA)$ and $S(BBB)$ are the trivial instances of coskewness. Similarly, Kurtosis is the trivial case of coskewness; e.g., $K(AAAA)$ and $K(BBBB)$ are the trivial instances of cokurtosis. In regard to this scatterplot (this two-variable application), how many nontrivial coskewness measures apply; are they positive or negative; and how many nontrivial cokurtosis measures exist?

A. There are two nontrivial coskewness measures and they are negative (aka, negative coskewness); there are three nontrivial cokurtosis measures

B. There are three nontrivial coskewness measures and they are negative (aka, negative coskewness); there are four nontrivial cokurtosis measures

C. There are two nontrivial coskewness measures and they are positive (aka, positive coskewness); there are five nontrivial cokurtosis measures

D. There are three nontrivial coskewness measures and they are positive (aka, positive coskewness); there are six nontrivial cokurtosis measures

答案:A
解析:
A. True: There are two nontrivial coskewness measures and they are negative (aka, negative coskewness); there are three nontrivial cokurtosis measures
When there are two variables, there are two non-trivial coskewness moments and three non-trivial cokurtosis moments. In regard to skew, the cross-moments are: $S(AAA)$, $S(AAB)$, $S(ABB)$, and $S(BBB)$ but $S(AAA)$ and $S(BBB)$ are the (trivial) univariate skewness moments. In regard to cokurtosis, the nontrivial cross-moments are $K(AAAB)$, $K(AABB)$, and $K(ABBB)$. Visually, the pair has negative coskewness because their negative returns tend to occur in the same period (see lower-left quadrant of the scatterplot where the relationship is fairly linear), but their positive returns tend to occur in different periods (see upper-right quadrant where the relationship is more scattered). If the funds were combined into one portfolio, their negative coskewness reflects their higher risk when returns are negative (as opposed to their somewhat diversified dynamic when returns are positive).
The spreadsheet below is located if you want to take a deeper dive (but further understanding is NOT necessary for the exam!): https://www.dropbox.com/s/i7bx2iewfrezgph/t2-20-13-3-sample-moments-3rd-coskewness.xlsx

n =	20						
			Coskew		CoKurt		
	Fund A	Fund B	S(AAB)	S(ABB)	K(AAAB)	K(AABB)	K(ABBB)
1	-6.0%	-8.1%	-0.0003	-0.0004	0.0000	0.0000	0.0000
2	-8.7%	-9.6%	-0.0008	-0.0008	0.0001	0.0001	0.0001
3	-7.1%	-6.1%	-0.0003	-0.0003	0.0000	0.0000	0.0000
4	-4.0%	-6.3%	-0.0001	-0.0002	0.0000	0.0000	0.0000
5	-5.0%	-6.4%	-0.0002	-0.0002	0.0000	0.0000	0.0000
6	-2.4%	-4.2%	0.0000	0.0000	0.0000	0.0000	0.0000
7	-3.3%	-2.2%	0.0000	0.0000	0.0000	0.0000	0.0000
8	-1.7%	-3.0%	0				00
9	-1.0%	-2.0%	0				00
10	1.3%	-1.1%	0				00
11	7.7%	0.4%	0				00
12	9.6%	1.5%	0				00
13	4.9%	1.8%	0				00
14	3.1%	2.0%	0				00
15	9.0%	3.9%	0				00
16	2.0%	5.0%	0				00
17	3.5%	5.2%	0				00
18	4.6%	5.8%	0.0001	0.0002	0.0000	0.0000	0.0000
19	0.5%	9.4%	0.0000	0.0000	0.0000	0.0000	0.0000
20	1.7%	9.1%	0.0000	0.0001	0.0000	0.0000	0.0000
Average	0.4%	-0.2%					
Sample Variance	0.3%	0.3%					
Sample StdDev	5.3%	5.6%					

	S(AAB)	S(ABB)	K(AAAB)	K(AABB)	K(ABBB)
cross-central moments	-0.0001	-0.0001	0.0000	0.0000	0.0000
standardized	-0.411	-0.457	1.634	1.387	1.423

考察: Quantitative Analysis　　**重要程度:** [■■□]
科目: Quantitative Analysis

67. Ralph creates a portfolio with two positions: a short call and a short put with identical maturities on the same stock that has a current price, $S(0) = \$20.00$. His short call with a strike price, $K_1 = \$22.00$ has a premium, $c_1 = \$1.75$. His short put with a strike price, $K_2 = \$18.00$ has a premium, $p_2 = \$1.25$. Both options are out-of-the-money by the same amount of \$2.00. Recall that option profit is equal to the payoff net of the premium and we (typically) ignore the impact of discounting. On the future date with both options mature, under which future stock price scenario is Ralph's trade profitable?

A. If the stock price falls within \$15.00 and \$25.00

B. His portfolio will always generate a small profit

C. If the stock price is less than \$18.00 or greater than \$22.00

D. If realized volatility increases; i.e., is higher on the future date

答案:A
解析:
Option A is correct. If the stock price falls within \$15.00 and \$25.00.
This portfolio is a trading strategy called a short strangle or top vertical combination which is a variation on the more common straddle. In a straddle, the investor buys a (European) call and put with identical strike prices and maturities. This is not a directional trade but rather a sort of volatility trade because it gains if the stock prices increase or decrease dramatically. The strangle is a variation because in the strangle the call's strike price is higher than the put's strike price. Writing the strangle, as illustrated in this question, is a short volatility trade because it is exposed to losses if the stock price increases or decreases greatly. Specifically, the initial cash inflow (the two premiums) is $\$1.75 + \$1.25 = \$3.00$. Therefore breakeven occurs when the payoff is \$3.00 which occurs when the stock price drops below \$15.00 (because the exercised put will cost \$3.00 at this price) or when the stock price increases above \$25.00 (because the exercised call will cost \$3.00 at this price). This short strangle generates a net profit if the stock price falls within $\{\$15.00, \$25.00\}$.

In regard to false (D), a straddle purchase (aka, bottom straddle) is a long volatility strategy, and a straddle write (aka, top straddle) is a short volatility strategy. Because the strike prices are different, the combination in this question is a short strangle and is similar to the straddle write because is a short volatility trade: it profits if the realized volatility is less than the initial implied volatility, but experiences a loss if the realized volatility is greater than the initial implied volatility.

考察: Finance: Options Strategies **重要程度:** [■■□]

科目: Quantitative Analysis

68. Sally owns a non-dividend-paying stock that is currently trading at \$60.00. As a hedge, she buys an out-of-the-money three-month European put option on the stock with a fixed strike price of \$57.00, which is a 5.0% discount to the current stock price. The stock's volatility is 36.0% per annum. The riskfree rate is 4.0%. According to the Black-Scholes-Merton model, what is the value of the put?

A. \$1.55

B. \$2.63

C. \$3.89

D. \$4.40

答案:B

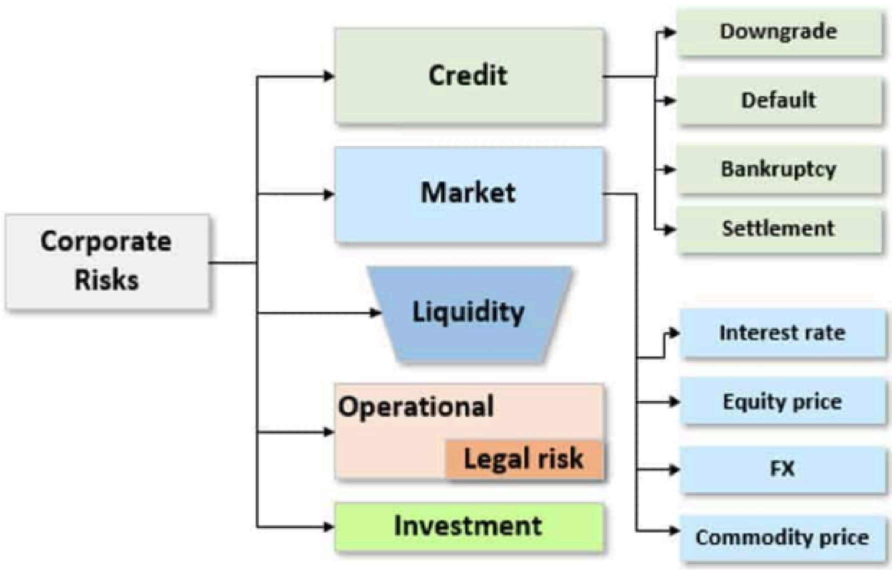
解析:

B 选项正确。 Per the BSM option pricing model, the value of a put is given by: $p = K \times \exp(-rT) \times N(-d_2) - S(0) \times N(-d_1)$. If we use the rounded lookup values, then $p = \$57.00 \times \exp(-0.040 \times 3/12) \times 0.4013 - \$60.00 \times 0.336 = \$2.630$. By design, this is also the exact value of the put (i.e., with exact normal Z value), per $c = \$57.00 \times \exp(-0.040 \times 3/12) \times 0.4011 - \$60.00 \times 0.334 = \$2.630$.

考察: Finance: Option Pricing **重要程度:** [■■□]

科目: Quantitative Analysis

69. Galaxy Financial is a large bank whose risk department developed the following risk typology after surveying key stakeholders within the bank:



Which of the following is most likely TRUE about their typology?

A. This typology is incorrect because it does not comport with any of the typologies approved by regulators, including Basel

B. If the typology is good then any and all newly identified risks should be slotted into an existing risk type within the typology

C. Although not illustrated directly here, many of the risk types interact with one another so that risk might flow or cascade from one type to another

D. It is incorrect to locate Liquidity risk as a top-level risk (i.e., alongside Credit and Market risk) because it should be a sub-risk to operational risk

答案:C

解析:

Option C is correct. Although not illustrated in the typology, many of the risk types interact with one another so that risk might flow from one type to another

As GARP explains, "The risk types interact with one another so that risk flows. During a severe crisis, for example, risk can flow from credit risk to liquidity risk to market risk, (which was the case during the global financial crisis of 2007–2009). The same can occur within an individual firm: the fat finger of an unlucky trader (operational risk) creates a dangerous market position (market risk) and potentially ruins the standing of the firm (reputational risk). That is why a sophisticated understanding of risk types and their interactions is an essential building block of risk management."[1]

In regard to (A), (B), and (D), each is FALSE

- In regard to false (A), we cannot deem the typology incorrect, especially without further information, as there is no single set of so-called correct typologies; nor do regulators qualify banks'typologies.
- In regard to false (B), the risk typology is dynamic, not static. New risks (e.g., cyber risk) create new types, or subdivide current types. Explains GARP, "Risk typologies must be flexible because new risks are always emerging."[1]
- In regard to false (D), consistent with flexibility, liquidity risk might be top-level, or often, it as sub-risk to Market Risk. Interestingly, in 2020 GARP introduced "Liquidity and Treasury Risk Measurement and Management"as a new top-level topic in Part 2 of the FRM. Previously Liquidity Risk was somewhat embedded into the Operational Risk Topic.

[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions, 10/2019

考察: Foundations of Risk Management　　**重要程度:** [■■□]

科目: Quantitative Analysis

70. An asset has a one-month volatility of 6.0%. If the asset's returns are i.i.d., the asset has a three-month volatility of $6.0\% \times \sqrt{3/1} = 10.4\%$ and a one-year volatility of $6.0\% \times \sqrt{12/1} = 20.8\%$. In general, by convention (and to avoid confusion), we should prefer the annualized measure: interest rate and volatility inputs and outputs are generally expressed in per annum terms. However, let's assume we are interested in the three-month value-at-risk (VaR) of this asset. If we assume that the asset's arithmetic returns, $[S(t) - S(t-1)]/S(t-1)$, have a normal distribution and i.i.d., then per the square root rule (SRR) the 95.0% confident three-month relative VaR of this asset is $6.0\% \times \sqrt{3/1} \times 1.645 = 17.10\%$; the "relative" adjective signifies that we are excluding the positive expected return which would mitigate this worst expected loss.

Now let's violate the i.i.d. assumption. Specifically, let's retain the "identical" but edit the "independence" to make an assumption of +0.50 auto- or serial-correlation between the returns; i.e., if the returns are correlated, they cannot be independent and i.i.d. is not satisfied such that we cannot rely on the square root rule (SRR). Now we are interested in the three-period (i.e., three-month) VaR based on a three-month volatility but where the variable (log return) has a positive correlation (with itself) of 0.50. Hint: per variance properties $\sigma^2(X+Y+Z) = \sigma^2(X+Y) + \sigma^2(Z) + 2 \times \mathrm{COV}(X+Y, Z)$ or equivalently $\sigma^2(R1+R2+R3) = \sigma^2(R1+R2) + \sigma^2(R3) + 2 \times \mathrm{COV}(R1+R2, R3)$.

If each one-month return has a standard deviation of 6.0% and the serial correlation between returns is +0.50, which is **nearest** to the three-month 95.0% normal relative VaR?

A. 17.1%

B. 24.2%

C. 29.6%

D. 51.5%

答案:B

解析:

B 选项正确。 Taking the $\sigma^2(R1+R2+R3) = \sigma^2(R1+R2) + \sigma^2(R3) + 2 \times \mathrm{COV}(R1+R2, R3)$, and extending:

Because $2 \times \text{COV}(R1+R2, R3) = 2 \times [\text{COV}(R1, R3) + \text{COV}(R2, R3)] = 2 \times \text{COV}(R1, R3) + 2 \times \text{COV}(R2, R3)$, and plugging back in:
$\sigma^2(R1+R2+R3) = \sigma^2(R1) + \sigma^2(R2) + \sigma^2(R3) + 2 \times \text{COV}(R1, R2) + 2 \times \text{COV}(R1, R3) + 2 \times \text{COV}(R2, R3)$; this is the three-variable variance formula
Per the "identical" assumption, $\sigma^2(R1) = \sigma^2(R2) = \sigma^2(R3) = 6.0\%^2$; and further $\rho(R1, R2) = \rho(R1, R3) = \rho(R2, R3) = 0.50$ such that:
$\sigma^2(R1+R2+R3) = 3 \times 6\%^2 + 3 \times 2 \times 6\% \times 6\% \times 0.5 = 0.02160$ and $\sigma(R1+R2+R3) = \sqrt{0.02160} = 14.70\%$
The three-month 95.0% normal relative VaR is given by $14.70\% \times 1.645 = 24.18\%$.
考察: Quantitative Analysis **重要程度:** [■■□]
科目: Quantitative Analysis

71. Joe is a 29-year-old single male who prefers to invest in a fund because he has neither the time nor interest to spend researching and selecting individual stocks. His financial advisor Sally is experience in the following fund categories: closed- and open-end mutual funds, hedge funds, and index funds. When Sally asks for Joe's preferences, his only strongly felt opinion is that he prefers to avoid paying high expenses or fees (reading the financial news gives him the impression that he should seek low fees because fees are trending toward zero). Which of the following is she MOST LIKELY to recommend to Joe?

A. An index fund (or exchange-traded fund, ETF) to achieve diversification with low fees

B. An active closed-end mutual fund that trades below its net asset value (NAV) because it will outperform as it pulls to maturity

C. A fund of hedge funds (FOHF) in order to overcome the principal-agent problem and ensure alignment between Joe's interest and the managers

D. An open-end mutual fund that has beaten the market in the prior three consecutive years because mutual fund performance tends to be persistent

答案:A
解析:
A 选项正确。 An index fund (or exchange-traded fund, ETF) to achieve diversification while controlling fees
I regard to false (B), (C) and (D), please note:

- The closed-end fund will not necessarily "pull to par": closed-end fund tend to trade at a discount
- The fund of hedge funds has two layers of fees (notoriously expensive!) and will not meet Joe's preference to keep fees low
- Past performance is no guarantee of future performance, and unfortunately, mutual fund performance tends not to be persistent. According to GARP (see "Mutual Fund Research"), actively managed funds on average do not outperform the market; and research has found that *persistence* is an elusive, uncommon feature among the vast majority of mutual funds.

考察: Finance: Investment Funds **重要程度:** [■■□]
科目: Quantitative Analysis

72. Given a universe of 500 stocks and an initial assumption for the risk-free rate, an analyst generates the efficient frontier. She assumes the mean-variance framework. The investment committee compliments her analysis and makes a request: they ask her to decrease the risk-free rate (the rate at which an investor can both borrow and lend) but otherwise make no changes (aka, ceteris paribus) to the expected returns or variance matrix for the equities. Which of the following is most likely to be correct?

A. The expected return on optimal portfolios with low risk & high risk will both decrease

B. The expected return on optimal portfolios with low risk will decrease, and the expected return on optimal portfolios with high risk will increase

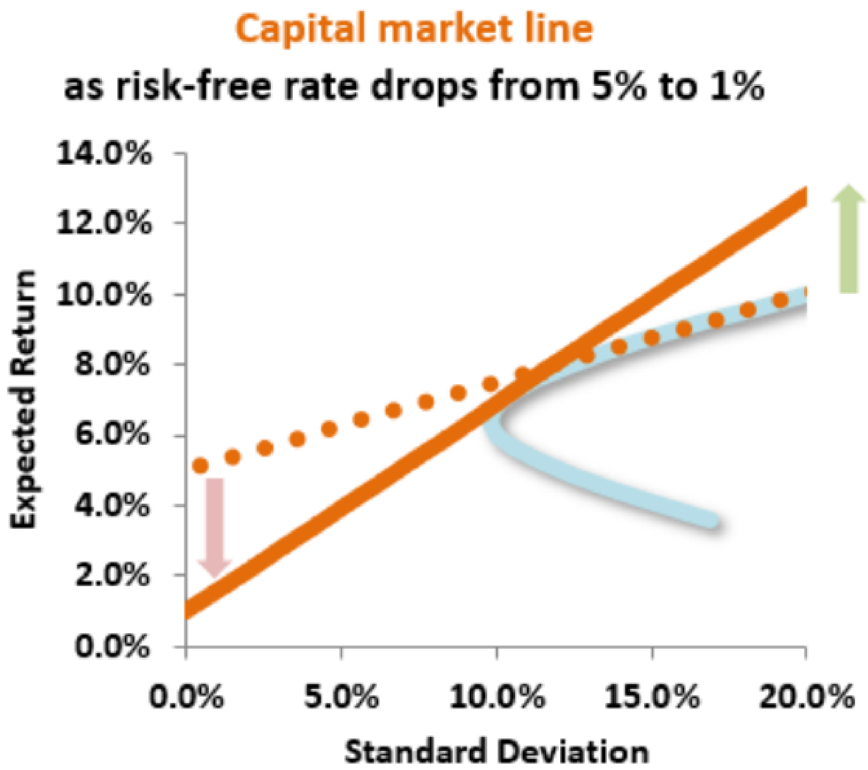
C. The expected return on optimal portfolios with low risk will increase, and the expected return on optimal portfolios with high risk will decrease

D. The expected return on optimal portfolios with low risk & high risk will both increase

答案:B

解析:

B 选项正确。 True: The expected return on optimal portfolios with low risk will decrease, and the expected return on optimal portfolios with high risk will increase Observe the following diagram:



As the risk-free rate moves downward, the new efficient frontier line now has a lower intercept and higher slope −so investors who pursue a rational strategy of selecting the maximum return for a given level of risk will have lower returns if they opt for a very low-risk portfolio but higher returns if they choose to be more aggressive on the risk dimension (standard deviation).

考察: Finance: Portfolio Theory　　**重要程度:** [■■□]

科目: Quantitative Analysis

73. In a single-factor economy, each of the following portfolios (A, B, and C) is well-diversified:

Riskfree rate	3.0%
Volatility of market index, σ[M]	30.0%

Portfolio	Beta	E[R(i)]	σ[e(i)]
A	0.60	12.0%	10.0%
B	0.80	15.0%	25.0%
C	1.20	???	42.0%

You discover there is NOT an arbitrage strategy among these three portfolios. In this case, what should be the expected return of Portfolio (C)?

A. 13.3%

B. 16.3%

C. 18.5%

D. 21.0%

答案:D
解析:
D 选项正确。 All three portfolios must lie on the same SML such that Portfolio C's Treynor ratio must be the same as the others. Treynor(Portfolio A) = $(12.0\% - 3.0\%)/0.60 = 0.15$; Treynor(Portfolio B) = $(15.0\% - 3.0\%)/0.80 = 0.15$, also. Therefore, Treynor(Portfolio C) = $(R - 3.0\%)/1.20 = 0.15$ such that its return must be 21.0%.
考察: Finance　**重要程度:**
科目: Valuation and Risk Models

74. Let's assume two normal variables: $X \sim N(0, 3.0^2)$ and $Y \sim N(4.0, 9.0^2)$. Let V be a mixture distribution where X has a component weight of 80% and Y has a component weight of 20%. Let $W = 0.80 * X + 0.20 * Y$ with the assumption that X and Y are independent; that is, this W is the sum of independent, random variables. Which is more likely (the mixture or the convolution) to realize a negative outcome; that is, is $\Pr(V < 0) > \Pr(W < 0)$, or on the other hand, is $\Pr(W < 0) > \Pr(V < 0)$?

A. The mixture (V) has a greater probability of a negative outcome; i.e., $\Pr(V < 0) > \Pr(W < 0)$

B. The convolution (W) has a greater probability of a negative outcome; i.e., $\Pr(W < 0) > \Pr(V < 0)$

C. They have the same probability of realizing a negative outcome; i.e., $\Pr(W < 0) = \Pr(V < 0)$

D. It cannot be answered because it depends on correlation in the case of (W)

答案:A
解析:
Option A is correct. The mixture (V) has a greater probability of a negative value; i.e., $\Pr(V < 0) > \Pr(W < 0)$

- In regard to the mixture distribution, the $\Pr(V < 0) = 80\% * 50.0\% + 20\% * 32.84\% = 40.0\% + 6.567\% = 46.57\%$.
- In regard to the convolution (i.e., sum of random variables which happen to be independent per given assumption), W has mean of 0.80 and standard deviation of $\sqrt{0.80^2 * 3.0^2 + 0.20^2 * 9.0^2} = 3.0$ such that $\Pr(W < 0) = 39.49\%$.

考察: Quantitative Analysis　**重要程度:** [■■□]
科目: Quantitative Analysis

75. Customers of insurance companies do not like to see their premiums increase, obviously. At the same time, most customers care about the claims process and quality: if the insurance doesn't pay out when it is needed, so the cheapest policy is not always the best! In regard to an increase in the premium(s), which of the following is MOST LIKELY to be TRUE?

A. A whole life insurance premium increases in the middle of the policy because the insurance company's shareholders have demanded an increase in the dividend

B. A variable life insurance premium increases because new genetic information (i.e., family tree insights) reduces the policy holder's life expectancy

C. A health insurance premium increases due to a higher cost for physician services, new technologies and/or hospital upgrades

D. A health insurance premium increases because the policyholder develops unexpected health problems that were unanticipated when the policy was written

答案:C
解析:
Option C is correct. A health insurance premium increases due to a higher cost for physician services, new technologies, and/or hospital upgrades.
In regard to (A), (B) and (D), each is FALSE. Instead,

- The life insurance company cannot increase the premium during the policy; they can increase the premium when the policy renews
- The Affordable Care Act modified the treatment of pre-existing conditions (https://en.wikipedia.org/wiki/Pre-existing_cond The problem with pre-existing conditions is that the insurance company is exposed to adverse selection which limits their ability to properly pool risk. However, the development of a health problem subsequent to the policy's inception is precisely the situation that insurance is meant for!

考察: Finance: Insurance **重要程度:** [■■□]
科目: Quantitative Analysis

76. In the wake of the global financial crisis (GFC), stress testing has received prominent media attention. Siddique and Hasan explain that "stress testing has many approaches. For example, at many institutions, finance and treasury have had ownership of the budgeting process. Hence, centralized stress testing could be housed in such a central function. Other institutions have taken a more decentralized approach, wherein the central function has only acted as an aggregator. Effective stress testing does not need a given organisational form or approach. However, a given organisational form will require certain aspects to ensure effectiveness. For example, in a decentralized approach towards stress testing, consistency across the organisation becomes important, and the institution will need to find ways of ensuring that consistency."[1]
Each of the following statements is true about stress testing EXCEPT which is false?
[1] Siddique and Hasan, Stress Testing: Approaches, Methods, and Applications (London: Risk Books, 2013)

A. Stress testing is a non-statistical risk measure that may employ scenario analysis as one approach

B. An acceptable proxy for stress testing is to generate value at risk (VaR) with a series of progressively increasing confidence levels; e.g., 99.0%, 99.50%, 99.75%, 99.90%, and 99.99%

C. While backtesting validates model parameter(s) using a large number of actual, historical observations; stress testing on the other hand uses a small number of extreme movements in market variables

D. Unlike value at risk (VaR) which indicates neither the magnitude nor shape of extreme losses beyond the VaR threshold, stress testing DOES attempt to characterize the magnitude/shape of the extreme loss tail

答案:B
解析:
Option B is correct. False, or at least it is not necessarily true that high levels of VaR confidence are good proxies for stress testing. VaR might be based on historical observations that do not anticipate (or imagine) discontinuous shifts. Related, this is also false for the same reason that choice (D) is true. In regard to (A), (C), and (D) each is TRUE.
考察: Risk Management **重要程度:** [■■□]
科目: Quantitative Analysis

77. Suppose that there are two independent economic factors, F1 and F2. The risk-free rate is 1.0%, and all stocks have independent firm-specific components with a standard deviation of 25%. The following are well-diversified portfolios; e.g., Portfolio (A) has a beta sensitivity to factor the first factor, $\beta(F1)$, of 1.20 and an expected return of 13.0%:

Portfolio	β(F1)	β(F2)	E(R)
A	1.20	-0.30	13.00%
B	0.60	-0.40	5.00%

Which is the correct return-beta relationship in this economy?

A. $E[R(P)] = 1.0\% - \beta(F1) * 6.0\% - \beta(F2) * 4.0\%$

B. $E[R(P)] = 1.0\% - \beta(F1) * 5.0\% + \beta(F2) * 2.0\%$

C. $E[R(P)] = 1.0\% + \beta(F1) * 9.0\% + \beta(F2) * 3.0\%$

D. $E[R(P)] = 1.0\% + \beta(F1) * 12.0\% + \beta(F2) * 8.0\%$

答案:D
解析:
D 选项正确。 $E[R(P)] = 1.0\% + \beta(F1) * 12.0\% + \beta(F2) * 8.0\%$.
We need to solve for two equations with two unknowns:

- $0.13 = 0.01 + 1.20 * RP(1) - 0.30 * RP(2)$
- $0.05 = 0.01 + 0.60 * RP(1) - 0.40 * RP(2)$

One way is to double the second equation and subtract it from the first in order to eliminate RP(1):

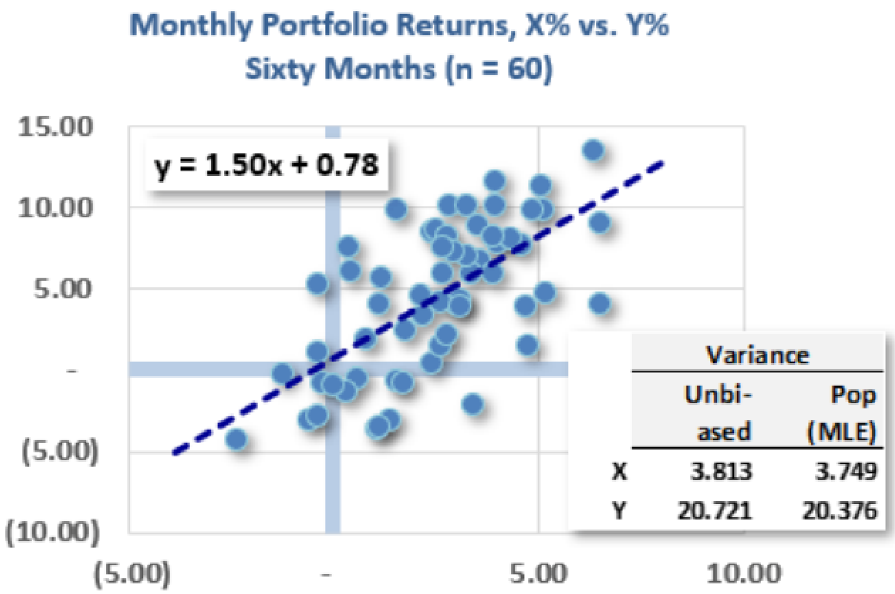
$$0.13 = 0.01 + 1.20 * RP(1) - 0.30 * RP(2)$$

$$-2 * [0.05 = 0.01 + 0.60 * RP(1) - 0.40 * RP(2)]$$

$$= 0.03 = -0.01 + 0.5 * RP(2) \rightarrow RP(2) = 0.04/0.5 = 0.08.$$

考察: Finance: Multifactor Models　　**重要程度:** [■■□]
科目: Quantitative Analysis

78. The scatterplot below plots sixty pairwise monthly returns over the last five years; each dot represents the return of Y (on the y-axis) and the return of X (on the x-axis) in a given month. Also shown is the ordinary least squares (OLS) regression line generated by Excel: $Y = 1.50 \times X + 0.78$. Finally, the univariate variances for each series are shown.



Which of the following is **nearest** to the correlation between X and Y? (Bonus question: if we flipped the axes, then what is the correlation?)

A. 0.28

B. 0.41

C. 0.64

D. 1.50

答案:C
解析:
C 选项正确。 True: 0.64 because $\rho(X,Y) = 1.50 \times \sqrt{3.813}/\sqrt{20.721} = 0.6435$
Because $\beta(Y,X) = \sigma(X,Y)/\sigma^2(X) = [\rho(X,Y)\sigma(X)\sigma(Y)]/\sigma^2(X) = \rho(X,Y)\sigma(Y)/\sigma(X)$ such that $\rho(X,Y) = \beta(Y,X)\sigma(X)/\sigma(Y)$; in this case, $\rho(X,Y) = 1.50 \times \sqrt{3.813}/\sqrt{20.721} = 0.643457$. Notice that we get the same result (!) if we use the biased variances, because the $(n-1)$ cancels: $\rho(X,Y) = 1.50 \times \sqrt{3.749}/\sqrt{20.376} = 0.643457$.
In regard to the bonus question, the correlation does not change if we flip the axes, although beta does change. If we flip the axes, the regression line is $y = 1.237 + 0.276x$, in which case $\rho(X,Y) = 0.276 \times \sqrt{20.376}/\sqrt{3.749} = 0.643457$.
考察: Quantitative Methods: Linear Regression **重要程度:** [■■□]
科目: Quantitative Analysis

79. The originate-to-distribute (OTD) banking model has both advantages and disadvantages. In regard to the OTD banking model, which of the following statements is TRUE?

A. OTD depresses the originating bank's reported return on equity (ROE)

B. OTD implies that investors should be aware of the adverse selection problem implied by information asymmetry

C. OTD has an inherent, unavoidable design bug and public relations problem: mezzanine and equity tranches absorb losses before senior tranches

D. OTD was banned after the global financial crisis (GFC 2007-08) such that it can no longer be conducted without incurring grave regulatory risk

答案:B
解析:
Option B is correct. OTD implies that investors should be aware of the adverse selection problem implied by information asymmetry
In regard to (A), (C), and (D), each is FALSE.

- In regard to (A), OTD shrinks the balance sheet and tends to increase ROE
- In regard to (C), subordination is one of the two key features of securitization (the other is credit risk transfer)
- In regard to (D), this greatly overstates the post-crisis reaction to OTD. As GARP explains, "The originate-to-distribute model played a role in the 2007–2008 crisis. Banks relaxed their mortgage lending standards so that the quality of the mortgages being originated declined. Despite this decline in quality, however, banks still managed to securitize them. In fact, they re-securitized mortgages by creating tranches from tranches. As defaults on the mortgages grew higher, losses were experienced by tranche holders. Unsurprisingly, for a period after the crisis, the originate-to-distribute model could not be used because it was not trusted by investors." — Source: Education, Pearson. Financial Markets and Products. Pearson Learning Solutions, 2020. VitalBook file.

考察: Financial Markets and Products **重要程度:** [■■□]
科目: Quantitative Analysis

80. Gigh Inc., a publicly traded company, had a wild swing today after OPEC cut production. Gigh ended the day with a -16% loss. Prior to this (most recently), the daily volatility was estimated to be 4.40% according to the exponentially weighted moving average (EWMA) model. The EWMA model had originally assumed a lambda, λ, of 0.70, but David wants to adjust the model's responsiveness to recent market conditions. David notices that his error is too high, so he wants to change the model to respond more quickly to changes in volatility, so he decides to decrease λ by 0.10. After David properly adjusts lambda, what is the new volatility estimate using EWMA?

A. $\sigma = 8.16\%$

B. $\sigma = 0.66\%$

C. $\sigma = 10.67\%$

D. $\sigma = 1.14\%$

答案:C
解析:
C 选项正确。$\sigma = 10.68\%$ A lower λ assigns more weight to the recent observations, making the EWMA respond more quickly to changes in volatility. Therefore, to make the model respond quicker to upticks in volatility David should decrease λ. Therefore,

$$\sigma_n^2 = \lambda * \sigma^2 + (1-\lambda) * (\text{return})^2$$

$$\text{New variance} = 0.6 * 4.40\%^2 + (1 - .6) * -16\%^2 = 0.0114016$$

$$\text{Volatility} = \text{SQRT}(0.0114016) = 10.68\%$$

考察: Quantitative Analysis **重要程度:** [■■□]
科目: Quantitative Analysis

81. Assume the riskfree rate is 3.0% while the market portfolio has a return of 13.0% (i.e., excess return is 10.0%) with volatility of 15.0%. Further, the following four portfolios are observed:

- Portfolio (A) has an expected return of 11.0% with volatility of 8.0%
- Portfolio (B) has an expected return of 7.0% with volatility of 20.0% and its correlation to the market is 0.30
- Portfolio (C) has a beta with respect to the market $\beta(C, M) = 1.50$ and its realized return of +20.0% implies an alpha of +2.0%
- Portfolio (D) has a beta with respect to the market $\beta(C, M) = 0.40$ and its realized return of +9.0% implies an alpha of +2.0%

Under the conditions of modern portfolio theory (MPT; mean-variance framework) and the capital asset pricing model (CAPM), each of the above portfolios is valid and plausible EXCEPT which is not possible?

A. Portfolio A

B. Portfolio B

C. Portfolio C

D. Portfolio D

答案:A
解析:
A 选项正确。 Portfolio A is unrealistic because it is "more efficient" than the Market portfolio: it has a Sharpe ratio of 1.0 versus the market portfolio's Sharpe ratio of $(13\% - 3\%)/15\% = 0.667$.
In regard to (B), (C), and (D), each is plausible

- Portfolio B has $\beta(B, M) = 0.30 * 20.0\%/15.0\% = 0.40$ with a CAPM expected return, $E(B) = 3.0\% + 0.40 * 10.0\% = 7.0\%$
- Portfolio C's Jensen's alpha, $\alpha(J, C) = 20.0\% - 3.0\% - 1.50 * 10.0\% = +2.0\%$
- Portfolio D's Jensen's alpha, $\alpha(J, D) = 9.0\% - 3.0\% - 0.40 * 10.0\% = +2.0\%$

考察: Finance: Portfolio Management **重要程度:** [■■□]
科目: Quantitative Analysis

82. In comparing and contrasting the law of large numbers (LLN) and the central limit theorem (CLT), each of the following statements is true EXCEPT which is false?

A. The LLN does not require independence but it does require normally distributed data

B. The CLT extends the LLN by requiring one additional assumption (the variance is finite)

C. LLN says that as the sample size increases ($n \to \infty$) the $\Pr(X_{\text{avg}} = \mu)$ approaches 100%

D. CLT says that as the sample size increases the distribution of the sample mean converges on the normal distribution $N(\mu, \sigma^2/n)$

答案:A
解析:
A 选项正确。 A. False. The inverse is true: the LLN requires independent (i.i.d.) random variables but is useful in large because (like the CLT) it makes no assumption about their distribution; it certainly does not require normally distributed random variables.
In regard to (B), (C) and (D), each is TRUE. Specifically:

- The Law of Large Numbers (LLN) says that as the sample size increases ($n \to \infty$) the $\Pr(X_{\text{avg}} = \mu)$ approaches 100%. This is the *essence* of the LLN. As GARP writes, "The Law of Large Numbers (LLN) establishes the large sample behavior of the mean estimator and provides conditions where an average converges to its expectation. The simplest LLN for iid random variables is the Kolmogorov Strong Law of Large Numbers." — 2020 FRM Part I: Quantitative Analysis, 10th Edition.

- The central limit theorem (CLT) extends the LLN by requiring one additional assumption (the variance is finite). While the LLN only requires a finite mean, the CLT adds one additional assumption that the variance also be finite.

- CLT says that as the sample size increases the distribution of the sample mean converges on the normal distribution $N(\mu, \sigma^2/n)$. This is the *essence* of the CLT: its magical superpower is that we require no knowledge of the random variable's distribution, yet if the sample is i.i.d. random (with finite mean and variance), then the sample mean is asymptotically normal.

考察: Quantitative Analysis **重要程度:** [■■□]
科目: Quantitative Analysis

83. For its most recent fiscal year, Acehouse Property-Casualty (APC) reports revenue of $300.0 million. Losses due to payouts (aka, loss and loss adjustment expense) were $213.0 million. Expenses (aka, underwriting expenses) were $69.0 million. Dividends paid were $6.0 million. Investment income was $9.0 million. What was the Operating ratio; aka, Combined ratio after dividends (CRAD) adjusted for Investment Income?

A. 7.0%

B. 23.0%

C. 93.0%

D. 106.0%

答案:C
解析:
C 选项正确。 TRUE: 93.0%. See below.

Revenue	$300.0	
Losses due to payouts / Loss ratio	$213.0	71.0%
Expenses / Expense ratio	$69.0	23.0%
Combined ratio	$282.0	94.0%
Dividends	$6.0	2.0%
Combined ratio after dividends (CRAD)	$288.0	96.0%
Investment Income	$9.0	3.0%
	$279.0	93.0%

考察: Finance: Insurance Ratios　　**重要程度:** [■■□]
科目: Quantitative Analysis

84. Scott is a risk manager for an investment manager. He is interested in estimating the potential loss the portfolio can expect over a specific time frame at a given confidence level (VaR) and the average of all potential losses beyond that potential loss (aka, expected shortfall, ES). Over the past 120 days, the portfolio's ten worst daily returns were the following:

-6.90%	-7.55%	-7.43%	-4.85%	-15.69%	-14.48%	-0.97%	-18.38%	-16.54%	-24.10%

Find the 1-day 5% VaR and ES for Scott (per Ch 2 Calculating and Applying VaR).

A. VaR 17.84%, ES 7.55%

B. VaR 7.43%, ES 16.12%

C. VaR 14.48%, ES 16.12%

D. VaR 14.48% ES 17.84%

答案:B
解析:
B 选项正确。 Step 1. Arrange values from worst to least bad day.

120	119	118	117	116	115	114	113	112	111
-24.10%	-18.38%	-16.54%	-15.69%	-14.48%	-7.55%	-7.43%	-6.90%	-4.85%	-0.97%

Step 2. Calculate the 5th percentage = $(5\% * 120)$ = 6th worst day + 1 = 7th worst day (per P2 Dowd). However, it should be noted that other chapters will show another way of calculating VaR, including using the 6th worst, 7th worst, or multiple interpolations, including using the median between the 6th and 7th worst.
Step 3. Identify the 7th worst day in our example it is -7.43%. This means that there is a 95% chance of having a loss of less than 7.43%.
Calculating ES Conditional is the average of losses that exceed VaR, not ES. However, ES is not ambiguous and is always the average of the 5% tail, which in this case is the 6 worst losses.

$$ES = \frac{(-24.1\% - 18.38\% - 16.54\% - 15.69\% - 14.48\% - 7.55\%)}{6} = 16.12\%$$

考察: Risk Management: VaR and ES　　**重要程度:** [■■□]
科目: Quantitative Analysis

85. A portfolio with a volatility of 30.0% has a Treynor measure of 0.080. The portfolio has a correlation of 0.50 with the market index which itself has a volatility of 20.0%. What is the portfolio's Sharpe measure?

A. 0.095

B. 0.200

C. 0.330

D. 0.475

答案:B
解析:
B 选项正确。 The beta(P, M) = correlation$(P, M) \times$ volatility(P)/volatility$(M) = 0.50 \times 30\%/20\% = 0.750$. Because Treynor = (Portfolio's excess return)/beta, the portfolio's excess return $= 0.080 \times 0.750 = 6.0\%$ and its Sharpe measure $= 6.0\%/30.0\% = 0.20$.
考察: Portfolio Management　　**重要程度:** [■■□]
科目: Quantitative Analysis

86. Darlene is a risk analyst who evaluates the creditworthiness of loan applicants at her financial institution. Her department is testing a new logistic regression model. If the model performs well in testing, it will be deployed to assist in the underwriting decision-making process. The training data is a sub-sample ($n = 800$) from the same LendingClub database used in reading. In the logistic regression, the dependent variable is a 0/1 for the terminal state of the loan being either zero (fully paid off) or one (deemed irrecoverable or defaulted). In the actual code, this dependent variable is labeled 'outcome'.
The following are the features (aka, independent variables) as given by their textual labels: Amount, Term, Interest_rate, Installment, Employ_hist, Income, and Bankruptcies. In regard to units in the database, please note the following: Amount is thousands of dollars ($000s); Term is months; Interest_rate is effectively multiplied by one hundred such that 7 equates to 7% or 0.070; Installment is dollars; Employment_hist is years; Income is thousand of dollars ($000); and Bankruptcies is a whole number $\{0, 1, 2, ...\}$.
The table below displays the logistic regression results:

Coeff_labels	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	−2.329	0.841	−2.769	0.006
Amount	0.000	0.000	1.339	0.181
Term	−0.041	0.027	−1.519	0.129
Interest rate	0.140	0.034	4.108	0.000
Installment	−0.003	0.003	−1.033	0.302
Employ Hist	−0.032	0.031	−1.025	0.305
Income	0.000	0.000	−0.937	0.349
Bankruptcies	0.457	0.230	1.991	0.046

In regard to this logistic regression, each of the following statements is true EXCEPT which is false?

A. A single additional bankruptcy increases the expected odds of default by almost 58 percent

B. If she requires significance at the 5% level or better, then two of the coefficients (in addition to the intercept) are significant

C. Each +100 basis points increase in the interest rate (e.g., from 8.0% to 9.0%), the expected default odds increase by about 15 percent

D. The key weakness of this regression is that Darlene cannot override the binary (yes/no) classification based on the standard $Z = 0.50$ threshold in order to avoid accepting too many applications

答案:D
解析:
D 选项正确。 In regard to (A), (B), and (C), each is TRUE.
考察: Quantitative Methods: Logistic Regression **重要程度:** [■■□]
科目: Quantitative Analysis

87. Consider the following four mutual funds and their respective behaviors:

- Fund A is an open-end fund that engages in "late trading" by accepting orders after 4 pm
- Fund B engages in "front running" by allowing its traders to make profitable trades in their own accounts ahead of the fund's large block trades
- Fund C engages in "directed brokerage" because it has an informal, reciprocal arrangement that guarantees the fund will use the brokerage for trades but only if the brokerage recommends its clients to the mutual fund
- Fund D engages in "timing the market" because, among its watchlist of 100 stocks, it buys (sells) whenever the price moves two standard deviations below (above) its 200-day moving average

Each of these behaviors is undesirable (or illegal) EXCEPT which is acceptable; i.e., which of the following statements is TRUE?

A. Fund A's behavior is permitted by the SEC

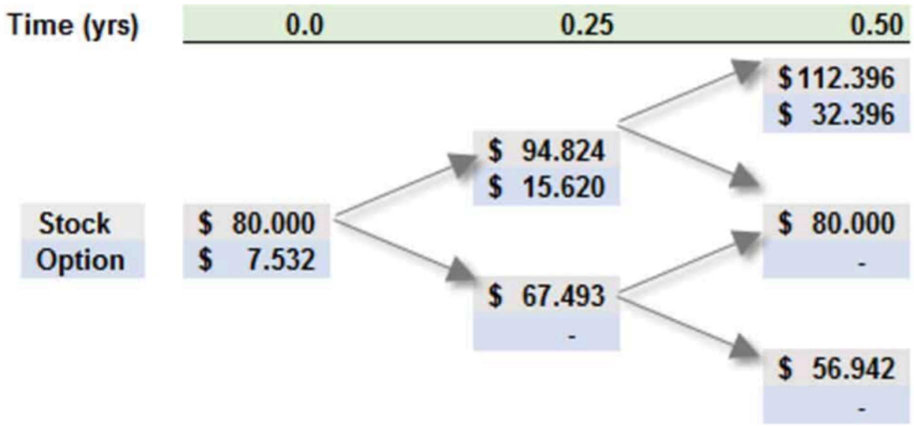
B. Fund B's behavior is legal

C. Fund C's behavior is encouraged by regulators

D. Fund D's behavior is permitted by regulators

答案:D
解析:
Option D is correct. Fund D's behavior (timing the market) is permitted by regulators, and is practiced by many investors. This timing is not to be confused with the "market timing" associated with buying/selling open-ended mutual fund shares at 4 pm when the actual values may be higher/lower than the slightly stale net asset value (NAV).
In regard to (A), (B), and (C), each is an undesirable trading behavior
考察: Finance: Mutual Funds **重要程度:** [■■□]
科目: Quantitative Analysis

88. Consider a six-month at-the-money (ATM) European call option on a non-dividend-paying stock with a current price of \$80.00. Peter the Risk Analyst has employed a two-step (i.e., three months per step) binomial model to price the option, as displayed below:



Peter's model matches the up and down movements to his estimate of the stock's prospective volatility, which he assumes is 34.0% per annum. The risk-free rate is 4.0%. Which of the following is nearest to the risk-neutral probability of the stock price going up in a single step?

A. 45.61%

B. 46.44%

C. 48.70%

D. 52.52%

答案:C

解析:

Option C is correct. 48.70%.

$u = \exp[\sigma\sqrt{\Delta t}] = \exp[0.340\sqrt{0.250}] = 1.18530$,

$d = \exp[-\sigma\sqrt{\Delta t}]$ or $1/d = 0.84366$, and

$p = [\exp(r\Delta t) - d]/(u - d) = [\exp(0.040 \times 0.25) - 0.84366]/(1.18530 - 0.84366) = 0.4870195$.

考察: Finance: Derivatives　　**重要程度:** [■■□]

科目: Quantitative Analysis

89. About risk reporting practices, the Committee says "[51.] Accurate, complete and timely data is a foundation for effective risk management. However, data alone does not guarantee that the board and senior management will receive appropriate information to make effective decisions about risk. To manage risk effectively, the right information needs to be presented to the right people at the right time. Risk reports based on risk data should be accurate, clear and complete. They should contain the correct content and be presented to the appropriate decision-makers in a time that allows for an appropriate response. To effectively achieve their objectives, risk reports should comply with the following principles. Compliance with these principles should not be at the expense of each other ..."[1]
Each of the following is a risk reporting principle EXCEPT which is not a risk reporting principle?
[1] "Principles for Effective Data Aggregation and Risk Reporting,"(Basel Committee on Banking Supervision Publication, January 2013)

A. Accuracy

B. Comprehensiveness

C. Clarity and usefulness

D. Manual workarounds

答案:D

解析:

D 选项正确。 Manual workarounds are "Employing human-based processes and tools to transfer, manipulate or alter data used to be aggregated or reported"and they are mentioned in risk data aggregation (39. Supervisors expect banks to document and explain all of their risk data aggregation processes whether automated or manual (judgment based or otherwise). Documentation should include an explanation of the appropriateness of any manual workarounds, a description of their criticality to the accuracy of risk data aggregation and proposed actions to reduce the impact."[1]

In regard to (A), (B) and (C) each is Risk Reporting practice. Basel Committee's Principles for effective risk data aggregation and risk reporting: "III. Risk reporting practices

- Principle 7: Accuracy –Risk management reports should accurately and precisely convey aggregated risk data and reflect risk in an exact manner. Reports should be reconciled and validated.
- Principle 8: Comprehensiveness –Risk management reports should cover all material risk areas within the organisation. The depth and scope of these reports should be consistent with the size and complexity of the bank's operations and risk profile, as well as the requirements of the recipients.
- Principle 9: Clarity and usefulness –Risk management reports should communicate information in a clear and concise manner. Reports should be easy to understand yet comprehensive enough to facilitate informed decision-making. Reports should include an appropriate balance between risk data, analysis and interpretation, and qualitative explanations. Reports should include meaningful information tailored to the needs of the recipients.
- Principle 10: Frequency –The board and senior management (or other recipients as appropriate) should set the frequency of risk management report production and distribution. Frequency requirements should reflect the needs of the recipients, the nature of the risk reported, and the speed at which the risk can change, as well as the importance of reports in contributing to sound risk management and effective and efficient decision-making across the bank. The frequency of reports should be increased during times of stress/crisis.
- Principle 11: Distribution –Risk management reports should be distributed to the relevant parties and while ensuring confidentiality is maintained."[1]

[1] "Principles for Effective Data Aggregation and Risk Reporting,"(Basel Committee on Banking Supervision Publication, January 2013)

考察: Risk Management　　**重要程度:** [■■□]

科目: Quantitative Analysis

90. Peter wants to model the following AR(2) time series: $Y(t) = 0.750 * Y(t-1) - 0.1250 * T(t-2) + e(t)$. He wonders if this AR(2) is stationary. He realizes that he can write this as a lag polynomial:

- $Y(t) = 0.750 * L - 0.1250 * L^2 + \varepsilon(t)$; first he expresses the model with lag operators
- $(1 - 0.750 * L + 0.1250 * L^2) * Y(t) = \varepsilon(t)$; then he isolates the lag polynomial
- $(1 - 0.50 * L)(1 - 0.250 * L) * Y(t) = \varepsilon(t)$; finally, he factors the polynomial

Which of the following statements about this AR(2) time series is true?

A. This AR(2) is non-stationary and Peter incorrectly factored the polynomial

B. This AR(2) is non-stationary because the sum of the coefficients is less than one

C. This AR(2) is stationary because the sum of the coefficients is less than one

D. This AR(2) is stationary because both roots of the characteristic equation are greater than one

答案:D

解析:

Option D is correct. This AR(2) is stationary because both roots of the characteristic equation are greater than one

Given this factored polynomial, $(1 - 0.50 * L)(1 - 0.250 * L) * Y(t)$, the roots solve for: $(1 - 0.50 * L) = 0$ such that $L = 2.0$; and $(1 - 0.250 * L) = 0$ such that $L = 4.0$.

If the absolute value of both roots are greater than unity (1.0), then the process is stationary.

考察: Quantitative Analysis　　**重要程度:** [■■□]

科目：Quantitative Analysis

91. The daily standard deviation of Bitcoin (BTC) is \$200.00 while the standard deviation of the futures contract price (on the same BTC) is \$340.00. The correlation between the two changes is 0.850. If the size of one futures contract is five Bitcoins (5 BTC), and the trader seeks to hedge the purchase of 100 Bitcoins (100 BTC), then how many contracts should be shorted to hedge?

A. One (1)

B. Three (3)

C. Ten (10)

D. Twenty five (25)

答案:C
解析:
C 选项正确。 The optimal hedge ratio, h^*, is given by $h^* = \beta(\Delta S/\Delta F) = \rho \cdot \sigma(S)/\sigma(F)$; in this case, $h^* = 0.850 \cdot \$200.00/\$340.00 = 0.50$.
The number of contracts to short is given by, $N^* = h^* \times Q_A/Q_F$; in this case, $N^* = 0.50 \cdot 100/5 = 10.0$.
考察：Risk Management: Hedging with Futures　　**重要程度：**[■■□]
科目：Quantitative Analysis

92. After Barbara wrote (i.e., sold) 100 put option contracts, the trade's position delta was +7,200. The put options were in-the-money as the stock price was \$70.00 while the options' strike price was \$100.00. Each contract was for 100 put options. The percentage gamma of each put option was +0.0120. If the stock price subsequently, immediately decreased by \$10.00 to \$60.00, which is **nearest** to an estimate of the trade's new position delta?

A. -3,500

B. +7,200 (unchanged)

C. +8,400

D. +11,900

答案:C
解析:
C 选项正确。 The percentage delta of the short put position was $7,200/-10,000 = -0.720$. Given a percentage gamma of +0.0120, a drop of \$10.00 in the stock price implies a change in delta given by $-\$10.00 \times 0.0120 = -0.120$, such that the new delta would be $-0.720 - 0.120 = -0.840$. Notice that, with the stock price drop, the put options become even more "in the money" and we do expect the percentage delta to drop (toward its limit at -1.0). Consequently, the new position delta is $-10,000 \times -0.840 = +8,400$.
考察：Valuation and Risk Models: Option Greeks　　**重要程度：**[■■□]
科目：Quantitative Analysis

93. The riskfree rate is 3.0% and the market portfolio's expected return is 10.0% (put another way, the market's excess expected return is 7.0%). Consider two portfolios:

- Portfolio A has a high volatility, $\sigma(A) = 50.0\%$ per annum, but its correlation to the market portfolio is only, $\rho(A, M) = 0.30$
- Portfolio B has a moderate volatility, $\sigma(B) = 30.0\%$ per annum, and its correlation to the market portfolio is, $\rho(B, M) = 0.70$

If both portfolios lie on the security market line (SML), and if Portfolio A has an expected return, $ER(A) = 7.20\%$, then what is the expected return of Portfolio B?

A. 5.100%

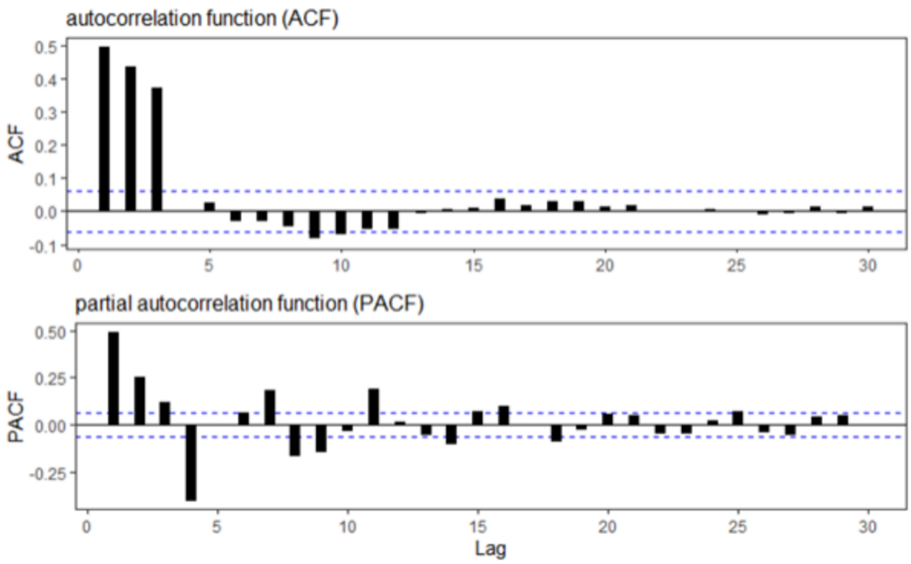
B. 7.900%

C. 8.880%

D. 9.540%

答案:C
解析:
C 选项正确。 True: 8.880%. If the both portfolios lie on the SML, then $ER(A) = Rf + \beta(A, M) * [ER(M) - Rf]$ and $ER(B) = Rf + \beta(B, M) * [ER(M) - Rf]$. We can infer Portfolio A's beta: since $7.20\% = 3.0\% + \beta(A, M) * 7.0\%$, $\beta(A, M) = 0.60$. And since $\beta(A, M) = \rho(A, M) * \sigma(A)/\sigma(M)$, we can infer that $\sigma(M) = [\rho(A, M) * \sigma(A)]/\beta(A, M) = [0.30 * 0.50]/0.60 = 0.25$. Then we can compute the beta of portfolio B: $\beta(B, M) = \rho(B, M) * \sigma(B)/\sigma(M) = 0.70 * 0.30/0.25 = 0.840$. The expected return of of portfolio B is given by $ER(B) = 3.0\% + 0.840 * 7.0\% = 8.880\%$.
考察: Finance: CAPM **重要程度:** [■■□]
科目: Financial Markets and Products

94. Below are plotted the autocorrelation function (ACF) and partial autocorrelation function (PACF) for a simulated time-series process. Please note that the horizontal dashed blue lines represent the 95.0% confidence interval.



Which of the following time-series models is **most likely** described by these ACF and PACF plots?

A. MA(1) with a coefficient of 1.3

B. MA(3) with coefficients of 0.4, 0.6, and 0.8

C. AR(2) with coefficients of 1.5 and 0.8

D. AR(3) with coefficients of 0.3, -1.4 and 0.7

答案:B
解析:
B 选项正确。 True: MA(3) with coefficients of 0.4, 0.6, and 0.8
The ACF cuts off (i.e., drops off sharply after) 3 lags. The PACF decays gradually. This suggests an MA(3) process. Explains GARP (emphasis ours): "The ACF is always zero for lags larger than q and the PACF is also complex. In general, the PACF of an MA(q) is non-zero at all lags ... The PACF oscillates and decays slowly towards zero ... These both [Figure 10.6] show the same pattern—one that is key to distinguishing MA models from AR models: The ACF cuts off sharply at the order of the

MA, whereas the PACF oscillates and decays towards zero over many lags. This is the opposite pattern from what is seen in Figure 10.2 and Figure 10.4, where the ACFs decayed slowly while the PACF cut off sharply." — 2020 FRM Part I: Quantitative Analysis, 10th Edition.

考察: Quantitative Analysis: Time Series　　**重要程度:** [■■□]

科目: Quantitative Analysis

95. Peter is a new analyst who has been asked to perform a valuation of a mortgage-backed security (MBS) using one of the firm's models. The pool is likely to be stripped into interest-only (IO) and principal-only (PO) strips. Each of the following statements is true EXCEPT which is false?

A. Interest rate uncertainty tends to reduce the value of IOs but increase the value of POs

B. His prepayment model will include an annualized prepayment rate that is likely to predict refinancings according to an incentive function

C. The prepayment rate in his model can be affected (in addition to interest rate levels) by average loan size, average loan age, season (aka, seasonality), and geographical location

D. His preferred model is a recombining tree because the cash flows are path independent because the permutation of up/down jumps prior to arrival at a future interest rate not is irrelevant

答案:D

解析:

D 选项正确。 False. Instead, he will want to model path dependence and, therefore, will prefer Monte Carlo Simulation. In regard to (A), (B), and (C), each is TRUE.

考察: Fixed Income: Mortgage-Backed Securities　　**重要程度:** [■■□]

科目: Quantitative Analysis

96. Trader Joe (who is unrelated to the awesome grocery store!) takes a long position in 100 out-of-the-money (OTM) put option contracts when the non-dividend-paying stock price is \$88.00 and its volatility is 28.0%; each contract is for 100 options. The strike price is \$80.00, and the option term is six months. The risk-free rate is 5.0%. The value of each option is \$4.38, and the position's value is $\$43,800.00$. With respect to these puts, the immediate percentage delta, $\Delta = -0.2550$ and gamma, $\Gamma = 0.0130$. If the stock price jumps by +\$7.00 to \$95.00, which is nearest to the position's value as approximated by delta and gamma; i.e., without a full re-pricing of the position?

A. $\$18,280$

B. $\$25,950$

C. $\$29,135$

D. $\$33,640$

答案:C

解析:

C 选项正确。 True: $\$29,135$

The delta-gamma approximated change is given by $\Delta S \cdot \Delta + 0.5 \cdot \Gamma \cdot (\Delta S)^2$; in this case of $\Delta S = +7.00$, estimated change $= 7.0 \cdot (-0.2550) + 0.5 \cdot (0.0130) \cdot 7.0^2 = -1.4665$. Each option is estimated to be worth $\$4.38 - \$1.4665 = \$2.9135$ and the position is estimated to be worth $\$29,135.00$.

In regard to false (B), $\$25,950$ is the estimated value based on only a delta approximation; i.e., if we exclude the gamma adjustment term.

The exact (re-priced with BSM) value is $\$28,841$, so the approximation is off by about 1.0%.

考察: Market Risk: The Greeks　　**重要程度:** [■■□]

科目: Quantitative Analysis

97. Which of the following is TRUE about the bank's audit function?

A. Internal audit is necessary because it is virtually impossible to rate (i.e., assign ratings to) the risk management function

B. The firm's operational risk management should report directly to the internal audit function (who will, therefore, oversee risk management)

C. Internal auditors are responsible for reviewing monitoring procedures, tracking the progress of risk management system upgrades, and affirming the efficacy of vetting processes

D. The assistance of internal audit should not be required to the risk governance function: a properly designed risk governance function should be able to ascertain compliance alone

答案:C
解析:
Option C is correct. Internal auditors are responsible for reviewing monitoring procedures, tracking the progress of risk management system upgrades, and affirming the efficacy of vetting processes.
In regard to (A), (B) and (D), each is FALSE.

- GARP recommends rating risk management practices: "A risk management function can be rated. This rating may be used internally or by third parties (e.g., rating agencies) that undertake comparative analyses of multiple enterprises. There is no one formula for excellence in risk management. Despite this, the rating of risk management practices would be instrumental in facilitating comparisons across an organization so that both the internal and external parties can benefit from such objective critiques," explains GARP (2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions)

- Risk management must be independent, even from internal audit: "Within the industry, there has been an active debate as to whether the audit function should have effective oversight of the firm's operational risk management. Note that the audit has a natural interest in the quality of internal controls. While subject to auditor review, however, the implementation of risk management must remain separate from the auditing function. As a basic principle, auditor independence from the underlying activity is essential to ensure confidence in any assurances or opinions rendered by the auditors to the board, and this applies equally to the risk management function and its associated processes. Unless this independence is maintained, conflicts of interest could compromise the quality of both risk management and audit activity and seriously jeopardize risk governance.", explains GARP.

- Internal audit is an essential complement to risk management: "Adherence to [the risk management] process can prevent the assumption of unbridled excessive risk. However, the risk governance function alone cannot ascertain compliance to the policies established by the board and external regulations. This is where the audit function comes in. It is incumbent upon the internal audit function to ensure the set-up, implementation, and efficacy of risk management/governance." explains GARP [1]

[1] 2020 FRM Part I: Foundations of Risk Management, 10th Edition. Pearson Learning Solutions
考察: Foundations of Risk Management　　**重要程度:** [■□□□□]
科目: Foundations of Risk Management

98. For each of three banks (Azure Bank, Breeze Bank, and Coastal Bank), Stella collected the following data on three features:

	Banks			Statistics		
	Azure	Breeze	Coastal	Mean	STDEV.S	STDEV.P
No. of customers (MM)	1.0	2.5	6.5	3.33	2.84	2.32
Size of loan book (BB)	$5.0	$12.5	$38.0	18.50	17.30	14.12
Number of branches	40	90	180	103.33	70.95	57.93

	Standardized (STDEV.S)		
	Azure	Breeze	Coastal
No. of customers (MM)	-0.821	-0.293	1.114
Size of loan book (BB)	-0.780	-0.347	1.127
Number of branches	-0.893	-0.188	1.081

Statistics displayed (top-right panel) include the mean, sample standard deviation (STDEV.S), and population standard deviation (STDEV.P). The standardized features (lower-left panel) are based on the sample standard deviation.
With respect to the standardized features (not the raw values), which of the following is **nearest** to the Euclidean distance between Breeze Bank and Coastal Bank?

A. 2.40

B. 4.15

C. 93.63

D. 119.50

答案:A
解析:
A 选项正确。 True: 2.40 is the Euclidean distance between the standardized features of Breeze Bank and Coastal Bank. The Euclidean distance between Breeze and Coastal is given by:

$$\sqrt{(-0.2930-1.1140)^2+(-0.3470-1.1270)^2+(-0.1880-1.0810)^2}=2.40056$$

In regard to (B), (C), and (D), each is false. Instead, they are alternative measures between Breeze Bank and Coastal Bank. Specifically:

- 4.15 is the Manhattan distance between their standardized features
- 93.63 is the Euclidean distance between their raw features
- 119.50 is the Manhattan distance between their raw features

考察: Quantitative Methods: Distance Metrics **重要程度:** [■■□]
科目: Quantitative Analysis

99. Among over-the-counter (OTC) interest rate derivatives, interest rate swap contracts are the most actively traded, according to BIS (source: https://www.bis.org/publ/qtrpdf/r_qt2212e.htm). Further, as Libor phases out, swaps will increasingly reference overnight rates such as SOFR and SONIA. About these modern overnight indexed swap (OIS) contracts, each of the following statements is true EXCEPT which is false?

A. In contrast to LIBOR, overnight rates such as SOFR and SONIA are nearly (or effectively) risk-free rates

B. The N-year swap rate is set equal to the average yield to maturity of bonds trading with N-year maturities

C. While LIBOR rates are known at the beginning of a period, overnight rates are known only at the end of each period

D. The confirmation specifies the swap's day count convention, payment exchange dates, payment calculation methodology, and the applicable holiday calendar

答案:B
解析:
B 选项正确。 False. The N-year swap rate is the market's N-year Par Yield because it's the N-year coupon rate that prices currently initiated swaps to par. Put another way, the swap rate is a par yield. On the other hand, the yield-to-maturity (aka yield) of current N-year bonds will vary as a function of their coupon rates.
In regard to (A), (C), and (D), each is TRUE.
考察: Financial Markets and Products: Interest Rate Swaps **重要程度:** [■■□]
科目: Quantitative Analysis

100. Analyst Patricia is analyzing the following four bonds:

- Bond A is a $100.00 face value bond with 7.0 years to maturity that pays a monthly coupon at a rate of 6.0% per annum and offers a yield of 5.0% per annum (with monthly compound frequency)

- Bond B is a \$100.00 face value bond with 10.0 years to maturity that pays a semi-annual coupon at a rate of 4.0% per annum and offers a yield of 5.0% per annum (with semi-annual compound frequency)
- Bond C is a \$100.00 face value bond with 10.0 years to maturity that pays an annual coupon at a rate of 7.0% per annum and offers a yield of 6.0% per annum (with annual compound frequency)
- Bond D is a \$1,000.00 face value zero-coupon bond with 30.0 years to maturity that offers a yield (aka, yield to maturity) of 8.0% per annum with semi-annual compound frequency

In terms of their current theoretical prices, which bonds are, respectively, the cheapest and most expensive (among the four)?

A. Bond A is the cheapest and Bond D is the most expensive

B. Bond B is the cheapest and Bond C is the most expensive

C. Bond C is the cheapest and Bond A is the most expensive

D. Bond D is the cheapest and Bond B is the most expensive

答案:B
解析:
Option B is correct. Bond B is the cheapest (at \$92.21) and Bond C is the most expensive (at \$107.36). The following are the theoretical prices:

- Bond A: $7 * 12 = 84$ N, $5/12 = 0.4167$ I/Y, $0.06 * 100/12 = 0.50$ PMT, 100 FV and CPT PV [+/-] = \$105.90
- Bond B: $10 * 2 = 20$ N, $5/2 = 2.5$ I/Y, $0.04 * 100/2 = 2.0$ PMT, 100 FV and CPT PV [+/-] = \$92.21
- Bond C: $10 * 1 = 10$ N, $6/1 = 6.0$ I/Y, $0.07 * 100/1 = 7.0$ PMT, 100 FV and CPT PV [+/-] = \$107.36
- Bond D: $30 * 2 = 60$ N, $8/2 = 4.0$ I/Y, 0 PMT, 100 FV and CPT PV [+/-] = \$95.06

考察: Fixed Income **重要程度:** [■■□]
科目: Quantitative Analysis

101. Alex is a junior analyst in the operational risk department at Global Trust Bank. His manager Sarah asks him to classify recent operational risk events according to Basel Committee's seven event categories. Which operational risk is correctly categorized?
Alex 是 Global Trust Bank 运营风险部门的一名初级分析师。他的经理 Sarah 要求他根据巴塞尔委员会的七个事件类别对最近的运营风险事件进行分类。哪一种运营风险被正确分类?

A. Reuters reports that a senior trader at Global Trust Bank manipulated the LIBOR submission process for three years, resulting in \$5 million illicit gains. Alex categorizes this as business disruption and system failures.
Reuters 报道，Global Trust Bank 的一名高级交易员连续三年操纵 LIBOR 提交过程，非法获利 500 万美元。Alex 将此事归类为业务中断和系统故障。

B. The bank's core trading platform experienced a four-hour outage during peak market hours, causing significant trading delays. Alex classifies this as execution, delivery, and process management.
银行的核心交易平台在高峰市场时段发生四小时宕机，导致重大交易延误。Alex 将此事归类为执行、交付和流程管理。

C. A severe earthquake damaged several branch offices in the coastal region, leading to \$20 million in property losses. Alex categorizes this as internal fraud.
一场严重的地震毁坏了沿海地区的一些分支机构，造成 2000 万美元的财产损失。Alex 将此事归类为内部欺诈。

D. Local media reveals that external hackers breached the bank's security system and stole sensitive client data, demanding \$3 million in ransom. Alex classifies this as external fraud.
当地媒体透露，外部黑客入侵了银行的安保系统，窃取了敏感的客户数据，并索要 300 万美元的赎金。Alex 将此事归类为外部欺诈。

答案:D
解析:
A 选项错误。

- LIBOR 操纵属于内部欺诈范畴，因为涉及员工故意违规获取个人利益。该事件包含了欺诈、故意违反规定等典型的内部欺诈特征，而不是业务中断和系统故障。

B 选项错误。

- 交易平台故障应归类为业务中断和系统故障，因为这涉及技术系统的运行中断。而执行、交付和流程管理通常指操作流程和交易处理中的人为错误。

C 选项错误。

- 地震造成的财产损失应归类为实物资产损坏，这类风险源于自然灾害或其他外部事件对银行实物资产的破坏，而不是内部欺诈行为。

D 选项正确。

- 外部黑客入侵和勒索属于典型的外部欺诈案例。这类事件由外部人员实施，目的是通过非法手段获取经济利益，符合外部欺诈的定义特征。

考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]
科目: Financial Markets and Products

102. Alex is a junior analyst in the operational risk department at Global Trust Bank. His manager Sarah asks him to classify recent operational risk events according to Basel Committee's seven event categories. Which operational risk is correctly categorized?
Alex 是 Global Trust Bank 运营风险部门的一名初级分析师。他的经理 Sarah 要求他根据巴塞尔委员会的七个事件类别对最近的运营风险事件进行分类。哪一种运营风险被正确分类?

A. Reuters reports that a senior trader at Global Trust Bank manipulated the LIBOR submission process for three years, resulting in \$5 million illicit gains. Alex categorizes this as business disruption and system failures.
Reuters 报道，Global Trust Bank 的一名高级交易员连续三年操纵 LIBOR 提交过程，非法获利 500 万美元。Alex 将此事归类为业务中断和系统故障。

B. The bank's core trading platform experienced a four-hour outage during peak market hours, causing significant trading delays. Alex classifies this as execution, delivery, and process management.
银行的核心交易平台在高峰市场时段发生四小时宕机，导致重大交易延误。Alex 将此事归类为执行、交付和流程管理。

C. A severe earthquake damaged several branch offices in the coastal region, leading to \$20 million in property losses. Alex categorizes this as internal fraud.
一场严重的地震毁坏了沿海地区的一些分支机构，造成 2000 万美元的财产损失。Alex 将此事归类为内部欺诈。

D. Local media reveals that external hackers breached the bank's security system and stole sensitive client data, demanding \$3 million in ransom. Alex classifies this as external fraud.
当地媒体透露，外部黑客入侵了银行的安保系统，窃取了敏感的客户数据，并索要 300 万美元的赎金。Alex 将此事归类为外部欺诈。

答案:D
解析:
A 选项错误。

- LIBOR 操纵属于内部欺诈范畴，因为涉及员工故意违规获取个人利益。该事件包含了欺诈、故意违反规定等典型的内部欺诈特征，而不是业务中断和系统故障。

B 选项错误。

- 交易平台故障应归类为业务中断和系统故障，因为这涉及技术系统的运行中断。而执行、交付和流程管理通常指操作流程和交易处理中的人为错误。

C 选项错误。

- 地震造成的财产损失应归类为实物资产损坏，这类风险源于自然灾害或其他外部事件对银行实物资产的破坏，而不是内部欺诈行为。

D 选项正确。

- 外部黑客入侵和勒索属于典型的外部欺诈案例。这类事件由外部人员实施，目的是通过非法手段获取经济利益，符合外部欺诈的定义特征。

考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]
科目: Financial Markets and Products

103. Alex is a junior analyst in the operational risk department at Global Trust Bank. His manager Sarah asks him to classify recent operational risk events according to Basel Committee's seven event categories. Which operational risk is correctly categorized?
Alex 是 Global Trust Bank 运营风险部门的一名初级分析师。他的经理 Sarah 要求他根据巴塞尔委员会的七个事件类别对最近的运营风险事件进行分类。哪一种运营风险被正确分类?

A. Reuters reports that a senior trader at Global Trust Bank manipulated the LIBOR submission process for three years, resulting in $5 million illicit gains. Alex categorizes this as business disruption and system failures.
Reuters 报道，Global Trust Bank 的一名高级交易员连续三年操纵 LIBOR 提交过程，非法获利 500 万美元。Alex 将此事归类为业务中断和系统故障。

B. The bank's core trading platform experienced a four-hour outage during peak market hours, causing significant trading delays. Alex classifies this as execution, delivery, and process management.
银行的核心交易平台在高峰市场时段发生四小时宕机，导致重大交易延误。Alex 将此事归类为执行、交付和流程管理。

C. A severe earthquake damaged several branch offices in the coastal region, leading to $20 million in property losses. Alex categorizes this as internal fraud.
一场严重的地震毁坏了沿海地区的一些分支机构，造成 2000 万美元的财产损失。Alex 将此事归类为内部欺诈。

D. Local media reveals that external hackers breached the bank's security system and stole sensitive client data, demanding $3 million in ransom. Alex classifies this as external fraud.
当地媒体透露，外部黑客入侵了银行的安保系统，窃取了敏感的客户数据，并索要 300 万美元的赎金。Alex 将此事归类为外部欺诈。

答案:D
解析:
A 选项错误。

- LIBOR 操纵属于内部欺诈范畴，因为涉及员工故意违规获取个人利益。该事件包含了欺诈、故意违反规定等典型的内部欺诈特征，而不是业务中断和系统故障。

B 选项错误。

- 交易平台故障应归类为业务中断和系统故障，因为这涉及技术系统的运行中断。而执行、交付和流程管理通常指操作流程和交易处理中的人为错误。

C 选项错误。

- 地震造成的财产损失应归类为实物资产损坏，这类风险源于自然灾害或其他外部事件对银行实物资产的破坏，而不是内部欺诈行为。

D 选项正确。

- 外部黑客入侵和勒索属于典型的外部欺诈案例。这类事件由外部人员实施，目的是通过非法手段获取经济利益，符合外部欺诈的定义特征。

考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]
科目: Financial Markets and Products

104. Alex is a junior analyst in the operational risk department at Global Trust Bank. His manager Sarah asks him to classify recent operational risk events according to Basel Committee's seven event categories. Which operational risk is correctly categorized?
Alex 是 Global Trust Bank 运营风险部门的一名初级分析师。他的经理 Sarah 要求他根据巴塞尔委员会的七个事件类别对最近的运营风险事件进行分类。哪一种运营风险被正确分类?

A. Reuters reports that a senior trader at Global Trust Bank manipulated the LIBOR submission process for three years, resulting in $5 million illicit gains. Alex categorizes this as business disruption and system failures.
Reuters 报道，Global Trust Bank 的一名高级交易员连续三年操纵 LIBOR 提交过程，非法获利 500 万美元。Alex 将此事归类为业务中断和系统故障。

B. The bank's core trading platform experienced a four-hour outage during peak market hours, causing significant trading delays. Alex classifies this as execution, delivery, and process management.
银行的核心交易平台在高峰市场时段发生四小时宕机，导致重大交易延误。Alex 将此事归类为执行、交付和流程管理。

C. A severe earthquake damaged several branch offices in the coastal region, leading to $20 million in property losses. Alex categorizes this as internal fraud.
一场严重的地震毁坏了沿海地区的一些分支机构，造成 2000 万美元的财产损失。Alex 将此事归类为内部欺诈。

D. Local media reveals that external hackers breached the bank's security system and stole sensitive client data, demanding $3 million in ransom. Alex classifies this as external fraud.
当地媒体透露，外部黑客入侵了银行的安保系统，窃取了敏感的客户数据，并索要 300 万美元的赎金。Alex 将此事归类为外部欺诈。

答案:D
解析:
A 选项错误。

- LIBOR 操纵属于内部欺诈范畴，因为涉及员工故意违规获取个人利益。该事件包含了欺诈、故意违反规定等典型的内部欺诈特征，而不是业务中断和系统故障。

B 选项错误。

- 交易平台故障应归类为业务中断和系统故障，因为这涉及技术系统的运行中断。而执行、交付和流程管理通常指操作流程和交易处理中的人为错误。

C 选项错误。

- 地震造成的财产损失应归类为实物资产损坏，这类风险源于自然灾害或其他外部事件对银行实物资产的破坏，而不是内部欺诈行为。

D 选项正确。

- 外部黑客入侵和勒索属于典型的外部欺诈案例。这类事件由外部人员实施，目的是通过非法手段获取经济利益，符合外部欺诈的定义特征。

考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]
科目: Financial Markets and Products

105. Alex is a junior analyst in the operational risk department at Global Trust Bank. His manager Sarah asks him to classify recent operational risk events according to Basel Committee's seven event categories. Which operational risk is correctly categorized?

Alex 是 Global Trust Bank 运营风险部门的一名初级分析师。他的经理 Sarah 要求他根据巴塞尔委员会的七个事件类别对最近的运营风险事件进行分类。哪一种运营风险被正确分类?

A. Reuters reports that a senior trader at Global Trust Bank manipulated the LIBOR submission process for three years, resulting in $5 million illicit gains. Alex categorizes this as business disruption and system failures.
Reuters 报道，Global Trust Bank 的一名高级交易员连续三年操纵 LIBOR 提交过程，非法获利 500 万美元。Alex 将此事归类为业务中断和系统故障。

B. The bank's core trading platform experienced a four-hour outage during peak market hours, causing significant trading delays. Alex classifies this as execution, delivery, and process management.
银行的核心交易平台在高峰市场时段发生四小时宕机，导致重大交易延误。Alex 将此事归类为执行、交付和流程管理。

C. A severe earthquake damaged several branch offices in the coastal region, leading to $20 million in property losses. Alex categorizes this as internal fraud.
一场严重的地震毁坏了沿海地区的一些分支机构，造成 2000 万美元的财产损失。Alex 将此事归类为内部欺诈。

D. Local media reveals that external hackers breached the bank's security system and stole sensitive client data, demanding $3 million in ransom. Alex classifies this as external fraud.
当地媒体透露，外部黑客入侵了银行的安保系统，窃取了敏感的客户数据，并索要 300 万美元的赎金。Alex 将此事归类为外部欺诈。

答案:D

解析:

A 选项错误。

- LIBOR 操纵属于内部欺诈范畴，因为涉及员工故意违规获取个人利益。该事件包含了欺诈、故意违反规定等典型的内部欺诈特征，而不是业务中断和系统故障。

B 选项错误。

- 交易平台故障应归类为业务中断和系统故障，因为这涉及技术系统的运行中断。而执行、交付和流程管理通常指操作流程和交易处理中的人为错误。

C 选项错误。

- 地震造成的财产损失应归类为实物资产损坏，这类风险源于自然灾害或其他外部事件对银行实物资产的破坏，而不是内部欺诈行为。

D 选项正确。

- 外部黑客入侵和勒索属于典型的外部欺诈案例。这类事件由外部人员实施，目的是通过非法手段获取经济利益，符合外部欺诈的定义特征。

考察: 偿付能力 II 对保险公司的资本要求　　**重要程度:** [■■□]

科目: Financial Markets and Products